

MATHEMATICS FOR NEUROSCIENCE

Student Workbook

Exercises with some worked examples

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Contents

1	Linear Algebra	2
1.1	Vectors and Vector Geometry	2
1.2	Matrices as Transformations	4
1.3	Systems of Equations and Matrix Inverses	6
1.4	Eigenvalues and Eigenvectors	9
1.5	Diagonalisation and Matrix Powers	13
2	Calculus	18
2.1	First-Order Separable ODEs	18
2.2	First-Order Linear ODEs with Forcing	21
2.3	The Deterministic Leaky Integrate-and-Fire Model	24
2.4	Second-Order Linear ODEs and Oscillations	29
2.5	Euler’s Method	32
2.6	Runge–Kutta Methods and Numerical Accuracy	37
2.7	Linear Systems of ODEs	41
2.8	Equilibria, Jacobians, and Local Stability	47
2.9	Parameter Changes and Bifurcations	51
3	Stochastic ODEs	56
3.1	Comparing ODEs and SDEs	56
3.2	Brownian Motion and Random Increments	61
3.3	Euler–Maruyama Simulation	66
3.4	Itô’s Formula and Change of Variables	71
3.5	The Stochastic Leaky Integrate-and-Fire Model	76

1 Linear Algebra

1.1 Vectors and Vector Geometry

Worked Example 1.1: Operations, length and orthogonality

Let

$$\mathbf{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

Calculate $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - \mathbf{v}$, $\|\mathbf{u}\|$, a unit vector in the direction of $\mathbf{u}$, and the angle between $\mathbf{u}$ and $\mathbf{v}$.

Solution. Vector operations are carried out component by component:

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 - 4 \\ 4 + 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix},$$

$$2\mathbf{u} - \mathbf{v} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix}.$$

The length of $\mathbf{u}$ is

$$\|\mathbf{u}\| = \sqrt{3^2 + 4^2} = 5.$$

Therefore, a unit vector in the direction of $\mathbf{u}$ is

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

Finally,

$$\mathbf{u} \cdot \mathbf{v} = 3(-4) + 4(3) = 0.$$

Since the dot product is zero, the vectors are perpendicular. Hence the angle between them is

$$\theta = 90^\circ.$$

Worked Example 1.2: Comparing neural population responses

The firing rates of three neurons are recorded under two stimuli:

$$\mathbf{r}_A = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{r}_B = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}.$$

Use the angle between the vectors to quantify how similar the two population-response patterns are.

Solution. The dot product is

$$\mathbf{r}_A \cdot \mathbf{r}_B = 5(4) + 2(1) + 1(3) = 25.$$

The lengths are

$$\|\mathbf{r}_A\| = \sqrt{5^2 + 2^2 + 1^2} = \sqrt{30}, \quad \|\mathbf{r}_B\| = \sqrt{4^2 + 1^2 + 3^2} = \sqrt{26}.$$

Using

$$\cos \theta = \frac{\mathbf{r}_A \cdot \mathbf{r}_B}{\|\mathbf{r}_A\| \|\mathbf{r}_B\|},$$

we obtain

$$\cos \theta = \frac{25}{\sqrt{30}\sqrt{26}} = \frac{25}{\sqrt{780}} \approx 0.895.$$

Therefore,

$$\theta \approx \cos^{-1}(0.895) \approx 26.5^\circ.$$

The angle is fairly small, so the two stimuli produce similar patterns across the three neurons, although the patterns are not identical.

Practice Questions

Question 1.1: Vector arithmetic

Let

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}.$$

Calculate:

- (a) $\mathbf{u} + \mathbf{v}$,
- (b) $\mathbf{u} - \mathbf{v}$,
- (c) $3\mathbf{u} + 2\mathbf{v}$,
- (d) $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$.

Question 1.2: Constructing a unit vector

Let

$$\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

Find a unit vector in the direction of $\mathbf{a}$, and verify explicitly that its length is one.

Question 1.3: Choosing a parameter for orthogonality

Let

$$\mathbf{p} = \begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}.$$

Find the value of k for which $\mathbf{p}$ and $\mathbf{q}$ are perpendicular.

Question 1.4: Angle between two vectors

Let

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Calculate the angle between the vectors. State whether the angle is acute, right or obtuse.

Question 1.5: Similarity of population codes

A population of three neurons responds to three stimuli with firing-rate vectors

$$\mathbf{r}_1 = \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{r}_2 = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{r}_3 = \begin{pmatrix} 0 \\ 3 \\ 6 \end{pmatrix}.$$

Use cosine similarity,

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|},$$

to decide whether $\mathbf{r}_2$ or $\mathbf{r}_3$ is more similar to $\mathbf{r}_1$. Interpret your result in terms of population-response patterns.

1.2 Matrices as Transformations**Worked Example 1.3: A horizontal shear**

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Calculate $A\mathbf{v}$, and describe geometrically how A transforms a general vector

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Solution. Multiplying the matrix by the vector gives

$$A\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1(2) + 1(1) \\ 0(2) + 1(1) \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

For a general vector,

$$A\mathbf{x} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ y \end{pmatrix}.$$

The vertical coordinate is unchanged, while the horizontal coordinate is increased by the vertical coordinate. This transformation is called a horizontal shear.

Worked Example 1.4: mixing neural signals

Suppose two underlying neural signals have amplitudes

$$\mathbf{s} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}.$$

Two recording channels measure linear mixtures of these signals:

$$\mathbf{y} = M\mathbf{s}, \quad M = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{4} & 1 \end{pmatrix}.$$

Calculate the recorded signal $\mathbf{y}$, and interpret the entries of the mixing matrix.

Solution. The recorded signal is

$$\mathbf{y} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{4} & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \end{pmatrix}.$$

Therefore,

$$\mathbf{y} = \begin{pmatrix} 1(6) + \frac{1}{2}(2) \\ \frac{1}{4}(6) + 1(2) \end{pmatrix} = \begin{pmatrix} 7 \\ \frac{7}{2} \end{pmatrix}.$$

The first recording channel contains the whole of the first neural signal and half of the second. The second recording channel contains one quarter of the first signal and the whole of the second.

The columns of M also show how each underlying signal contributes to the two recording channels:

$$M \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{4} \end{pmatrix}, \quad M \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}.$$

Practice Questions

Question 1.6: Applying a matrix transformation

Let

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

Calculate $A\mathbf{v}$.

Question 1.7: Identity and scaling transformations

Let

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}.$$

- Calculate $I\mathbf{u}$.
- Calculate $D\mathbf{u}$.
- Describe geometrically what the matrix D does to a general vector.

Question 1.8: The order of transformations

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- Calculate $B(A\mathbf{v})$.
- Calculate $A(B\mathbf{v})$.
- Are the two answers equal?
- Explain what this shows about the order in which matrix transformations are applied.

Question 1.9: Constructing a transformation matrix

A linear transformation T acts on the standard basis vectors as follows:

$$T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

(a) Write down the matrix A representing T .

(b) Use A to calculate

$$T \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Question 1.10: Rotation and preservation of geometry

The matrix

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

rotates vectors anticlockwise through an angle θ .

Let

$$\theta = \frac{\pi}{2}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

(a) Write down the numerical matrix R .

(b) Calculate $R\mathbf{v}$ and $R\mathbf{w}$.

(c) Verify that the lengths of both vectors are preserved.

(d) Verify that

$$(R\mathbf{v}) \cdot (R\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}.$$

(e) What does the final result tell you about the angle between the two vectors?

1.3 Systems of Equations and Matrix Inverses**Worked Example 1.5: Solving a system using a matrix inverse**

Consider the system

$$\begin{aligned} 2x + y &= 1, \\ 5x + 4y &= 4. \end{aligned}$$

Write the system in matrix form and solve it using a matrix inverse.

Solution. The system can be written as

$$A\mathbf{u} = \mathbf{b},$$

where

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 4 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

The determinant of A is

$$\det(A) = 2(4) - 1(5) = 3.$$

Since the determinant is non-zero, the inverse exists:

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 4 & -1 \\ -5 & 2 \end{pmatrix}.$$

Therefore,

$$\mathbf{u} = A^{-1}\mathbf{b} = \frac{1}{3} \begin{pmatrix} 4 & -1 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Hence

$$\mathbf{u} = \frac{1}{3} \begin{pmatrix} 4 - 4 \\ -5 + 8 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Thus

$$x = 0, \quad y = 1.$$

Substituting into the original equations verifies the result:

$$2(0) + 1 = 1, \quad 5(0) + 4(1) = 4.$$

Worked Example 1.6: Recovering neural sources

Suppose two underlying neural sources have unknown amplitudes s_1 and s_2 . Two recording channels measure

$$\begin{aligned} y_1 &= s_1 + s_2, \\ y_2 &= 2s_1 + 3s_2. \end{aligned}$$

The measured values are

$$y_1 = 7, \quad y_2 = 18.$$

Use a matrix inverse to recover the two source amplitudes.

Solution. The recording equations can be written as

$$M\mathbf{s} = \mathbf{y},$$

where

$$M = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} s_1 \\ s_2 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 7 \\ 18 \end{pmatrix}.$$

The determinant is

$$\det(M) = 1(3) - 1(2) = 1,$$

so M is invertible. Its inverse is

$$M^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}.$$

Therefore,

$$\mathbf{s} = M^{-1}\mathbf{y} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 18 \end{pmatrix} = \begin{pmatrix} 21 - 18 \\ -14 + 18 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Thus the recovered source amplitudes are

$$s_1 = 3, \quad s_2 = 4.$$

Checking,

$$3 + 4 = 7, \quad 2(3) + 3(4) = 18.$$

Practice Questions

Question 1.11: Writing and solving a system in matrix form

Consider the system

$$3x + y = 7,$$

$$2x - y = 3.$$

- (a) Write the system in the form

$$A\mathbf{u} = \mathbf{b}.$$

- (b) Calculate $\det(A)$.

- (c) Find A^{-1} .

- (d) Use the inverse to calculate x and y .

- (e) Verify your answer in the original equations.

Question 1.12: Calculating and checking an inverse

Let

$$M = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix}.$$

- (a) Calculate $\det(M)$.

- (b) Find M^{-1} .

- (c) Verify your result by calculating

$$M^{-1}M.$$

Question 1.13: A singular system

Consider the matrix

$$A_k = \begin{pmatrix} k & 2 \\ 3 & 6 \end{pmatrix}.$$

- (a) Find $\det(A_k)$.

- (b) Find the value of k for which A_k has no inverse.

- (c) For this value of k , consider the two systems

$$A_k \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}$$

and

$$A_k \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \end{pmatrix}.$$

Explain why the first system has infinitely many solutions, while the second system has no solution.

Question 1.14: Separating two sources

Two recording channels measure mixtures of two neural sources:

$$y_1 = 2s_1 + s_2,$$

$$y_2 = s_1 + 2s_2.$$

The observed recording is

$$\mathbf{y} = \begin{pmatrix} 8 \\ 7 \end{pmatrix}.$$

(a) Write the equations in the form

$$M\mathbf{s} = \mathbf{y}.$$

(b) Calculate M^{-1} .

(c) Recover s_1 and s_2 .

(d) Verify that the recovered sources reproduce the observed recording.

Question 1.15: Using one inverse to solve two systems

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Use a single calculation of A^{-1} to solve both systems

$$A\mathbf{x}_1 = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$$

and

$$A\mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Verify both answers by multiplying them by A .

1.4 Eigenvalues and Eigenvectors**Worked Example 1.7: Finding eigenvalues and eigenvectors**

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

Find the eigenvalues and corresponding eigenvectors of A . Verify each eigenpair.

Solution. The eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

We have

$$A - \lambda I = \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix},$$

so

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2.$$

Expanding,

$$\det(A - \lambda I) = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

Therefore, the eigenvalues are

$$\lambda_1 = 2, \quad \lambda_2 = 5.$$

For $\lambda_1 = 2$,

$$(A - 2I)\mathbf{v} = \mathbf{0},$$

so

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives

$$2x + y = 0.$$

Choosing $x = 1$ gives $y = -2$, so one eigenvector is

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

For $\lambda_2 = 5$,

$$(A - 5I)\mathbf{v} = \mathbf{0},$$

so

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives

$$-x + y = 0.$$

Choosing $x = 1$ gives $y = 1$, so one eigenvector is

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

To verify the first eigenpair,

$$A\mathbf{v}_1 = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Similarly,

$$A\mathbf{v}_2 = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Worked Example 1.8: Activity modes in a recurrent network

Consider a two-population recurrent network with update rule

$$\mathbf{r}_{n+1} = W\mathbf{r}_n,$$

where

$$W = \begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}.$$

Find the eigenvalues and eigenvectors of W , and interpret the two activity modes.

Solution. The eigenvalues satisfy

$$\det(W - \lambda I) = 0.$$

Thus

$$\det \begin{pmatrix} 0.8 - \lambda & 0.2 \\ 0.1 & 0.7 - \lambda \end{pmatrix} = 0.$$

Therefore,

$$(0.8 - \lambda)(0.7 - \lambda) - 0.02 = 0.$$

Expanding,

$$\lambda^2 - 1.5\lambda + 0.54 = 0,$$

which factors as

$$(\lambda - 0.9)(\lambda - 0.6) = 0.$$

Hence

$$\lambda_1 = 0.9, \quad \lambda_2 = 0.6.$$

For $\lambda_1 = 0.9$,

$$W - 0.9I = \begin{pmatrix} -0.1 & 0.2 \\ 0.1 & -0.2 \end{pmatrix}.$$

The equation

$$-0.1x + 0.2y = 0$$

gives $x = 2y$. One eigenvector is therefore

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 0.6$,

$$W - 0.6I = \begin{pmatrix} 0.2 & 0.2 \\ 0.1 & 0.1 \end{pmatrix}.$$

The equation

$$x + y = 0$$

gives the eigenvector

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The first mode has both neural populations active with the same sign. Each update scales this pattern by 0.9, so it decays relatively slowly.

The second mode represents a difference between the two populations. Each update scales this pattern by 0.6, so it decays more quickly.

Practice Questions

Question 1.16: Checking proposed eigenvectors

Let

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- (a) Calculate $A\mathbf{v}_1$.
- (b) Calculate $A\mathbf{v}_2$.
- (c) Show that both vectors are eigenvectors of A .
- (d) State the eigenvalue associated with each vector.

Question 1.17: Calculating two eigenpairs

Let

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}.$$

- (a) Form the characteristic equation

$$\det(A - \lambda I) = 0.$$
- (b) Find the two eigenvalues.
- (c) Find one eigenvector corresponding to each eigenvalue.
- (d) Verify both eigenpairs.

Question 1.18: Choosing a parameter

Let

$$A_k = \begin{pmatrix} 2 & k \\ 0 & 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Find the value of k for which $\mathbf{v}$ is an eigenvector of A_k . State the corresponding eigenvalue.

Question 1.19: Repeated transformation of an eigenvector

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

- (a) Verify that $\mathbf{u}$ is an eigenvector of A .
- (b) State its eigenvalue.
- (c) Without multiplying the matrix repeatedly, calculate

$$A^2\mathbf{u}, \quad A^3\mathbf{u}, \quad A^5\mathbf{u}.$$

- (d) Write a formula for $A^n\mathbf{u}$.

Question 1.20: Evolution of two activity modes

Consider

$$\mathbf{r}_{n+1} = W\mathbf{r}_n, \quad W = \begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}.$$

The matrix has eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with eigenvalues 0.9 and 0.6, respectively.

Suppose the initial activity is

$$\mathbf{r}_0 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

(a) Show that

$$\mathbf{r}_0 = \mathbf{v}_1 + \mathbf{v}_2.$$

(b) Calculate $\mathbf{r}_1$ using the eigenvectors and eigenvalues.

(c) Write an expression for $\mathbf{r}_n$.

(d) Which activity mode dominates for large n ?

(e) Explain what eventually happens to the total activity.

1.5 Diagonalisation and Matrix Powers

A matrix can be diagonalised when it has enough linearly independent eigenvectors to form an invertible matrix P . If the columns of P are eigenvectors of A , then

$$A = PDP^{-1},$$

where D is the diagonal matrix containing the corresponding eigenvalues.

Worked Example 1.9: Diagonalising a matrix and calculating a power

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

The eigenvalues and eigenvectors are

$$\lambda_1 = 2, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

and

$$\lambda_2 = 5, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Construct matrices P and D , verify that

$$A = PDP^{-1},$$

and use the diagonalisation to calculate A^4 .

Solution. Place the eigenvectors in the columns of P , in the same order as the eigenvalues in D :

$$P = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

The determinant of P is

$$\det(P) = 1(1) - 1(-2) = 3,$$

so

$$P^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

We can first verify that

$$AP = PD.$$

Indeed,

$$AP = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ -4 & 5 \end{pmatrix},$$

while

$$PD = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ -4 & 5 \end{pmatrix}.$$

Since $AP = PD$, multiplying on the right by P^{-1} gives

$$A = PDP^{-1}.$$

For a positive integer n ,

$$A^n = PD^nP^{-1}.$$

Therefore,

$$A^4 = P \begin{pmatrix} 2^4 & 0 \\ 0 & 5^4 \end{pmatrix} P^{-1}.$$

Thus

$$A^4 = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 16 & 0 \\ 0 & 625 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

Multiplying gives

$$A^4 = \frac{1}{3} \begin{pmatrix} 1266 & 609 \\ 1218 & 657 \end{pmatrix} = \begin{pmatrix} 422 & 203 \\ 406 & 219 \end{pmatrix}.$$

Worked Example 1.10: decomposing population activity into modes

Consider a two-population recurrent network

$$\mathbf{r}_{n+1} = W\mathbf{r}_n,$$

where

$$W = \begin{pmatrix} 0.7 & 0.2 \\ 0.2 & 0.7 \end{pmatrix}.$$

The initial activity is

$$\mathbf{r}_0 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Diagonalise W , and derive an expression for $\mathbf{r}_n$.

Solution. The vectors

$$\mathbf{v}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

are eigenvectors because

$$W\mathbf{v}_+ = \begin{pmatrix} 0.9 \\ 0.9 \end{pmatrix} = 0.9\mathbf{v}_+,$$

and

$$W\mathbf{v}_- = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} = 0.5\mathbf{v}_-.$$

Therefore,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

The inverse is

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

To express the initial activity in the eigenvector basis, calculate

$$P^{-1}\mathbf{r}_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Thus

$$\mathbf{r}_0 = 3\mathbf{v}_+ + 2\mathbf{v}_-.$$

After n updates,

$$\mathbf{r}_n = W^n \mathbf{r}_0 = 3(0.9)^n \mathbf{v}_+ + 2(0.5)^n \mathbf{v}_-.$$

Therefore,

$$\mathbf{r}_n = 3(0.9)^n \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2(0.5)^n \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Both modes decay because both eigenvalues have magnitude less than one. The common-activity mode $\mathbf{v}_+$ decays more slowly and therefore dominates the relative population pattern for large n .

Practice Questions

Question 1.21: Constructing a diagonalisation

Let

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- Verify that $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of A , and find their eigenvalues.
- Construct the eigenvector matrix P .
- Construct the diagonal eigenvalue matrix D .
- Calculate P^{-1} .
- Verify that

$$AP = PD$$

and hence that

$$A = PDP^{-1}.$$

Question 1.22: Finding a formula for A^n

Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

The matrix has eigenvalues and eigenvectors

$$\lambda_1 = 2, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and

$$\lambda_2 = 1, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

- (a) Construct P , D and P^{-1} .
- (b) Use diagonalisation to show that

$$A^n = \begin{pmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{pmatrix}.$$

- (c) Calculate A^5 .

Question 1.23: Decomposing a vector into eigenmodes

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

Its eigenpairs are

$$\lambda_1 = 2, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

and

$$\lambda_2 = 5, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Let

$$\mathbf{x}_0 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

- (a) Find constants c_1 and c_2 such that

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$$

- (b) Write an expression for

$$A^n \mathbf{x}_0.$$

- (c) Calculate $A^3 \mathbf{x}_0$.
- (d) Which eigenmode dominates for large n ?

Question 1.24: Growth and decay under repeated transformation

Let

$$B = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}.$$

The matrix has eigenvectors

$$\mathbf{v}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with eigenvalues $3/2$ and $1/2$, respectively.

Suppose

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

- Express $\mathbf{x}_0$ as a linear combination of $\mathbf{v}_+$ and $\mathbf{v}_-$.
- Write an expression for $\mathbf{x}_n = B^n \mathbf{x}_0$.
- Calculate $\mathbf{x}_4$.
- Which mode dominates for large n ?
- Does $\mathbf{x}_n$ approach zero, remain bounded or grow?

Question 1.25: Long-term population pattern

Consider the recurrent neural network

$$\mathbf{r}_{n+1} = W \mathbf{r}_n,$$

where

$$W = \begin{pmatrix} 0.75 & 0.15 \\ 0.05 & 0.65 \end{pmatrix}.$$

The matrix has eigenpairs

$$\lambda_1 = 0.8, \quad \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

and

$$\lambda_2 = 0.6, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The initial population activity is

$$\mathbf{r}_0 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}.$$

- Show that

$$\mathbf{r}_0 = \mathbf{v}_1 + \mathbf{v}_2.$$
- Write an expression for $\mathbf{r}_n$.
- Calculate $\mathbf{r}_3$.
- Which mode dominates the relative population pattern for large n ?
- What happens to the magnitude of the total activity?

2 Calculus

2.1 First-Order Separable ODEs

Worked Example 2.1: Exponential growth and decay

Suppose a population $N(t)$ satisfies

$$\frac{dN}{dt} = aN, \quad N(0) = N_0,$$

where a is a constant.

Solve the differential equation and describe how the sign of a affects the solution.

Solution. Separate the variables:

$$\frac{1}{N} dN = a dt.$$

Integrating both sides gives

$$\int \frac{1}{N} dN = \int a dt,$$

so

$$\ln |N| = at + C.$$

Exponentiating,

$$N = Ke^{at},$$

where K is a constant.

Using the initial condition,

$$N(0) = K = N_0.$$

Therefore,

$$\boxed{N(t) = N_0 e^{at}}.$$

If $a > 0$, the population grows exponentially.

If $a < 0$, the population decays exponentially towards zero.

If $a = 0$, the population remains constant:

$$N(t) = N_0.$$

To verify the solution,

$$\frac{d}{dt} (N_0 e^{at}) = a N_0 e^{at} = a N(t).$$

Worked Example 2.2: exponential neurotransmitter clearance

Suppose the concentration $C(t)$ of a neurotransmitter after release satisfies

$$\frac{dC}{dt} = -\frac{1}{\tau} C, \quad C(0) = C_0,$$

where $\tau > 0$ is the clearance time constant.

(a) Find $C(t)$.

(b) Let $C_0 = 8 \mu\text{M}$ and $\tau = 20 \text{ ms}$. Calculate $C(20 \text{ ms})$.

(c) Find the time at which the concentration has fallen to half its initial value.

Solution. Separate the variables:

$$\frac{1}{C} dC = -\frac{1}{\tau} dt.$$

Integrating,

$$\ln |C| = -\frac{t}{\tau} + K.$$

Therefore,

$$C(t) = Ae^{-t/\tau}.$$

Using $C(0) = C_0$ gives $A = C_0$, so

$$C(t) = C_0 e^{-t/\tau}.$$

At $t = 20$ ms, with $\tau = 20$ ms,

$$C(20) = 8e^{-20/20} = 8e^{-1}.$$

Thus

$$C(20) \approx 2.94 \mu\text{M}.$$

For the half-life, set

$$C(t_{1/2}) = \frac{C_0}{2}.$$

Then

$$\frac{C_0}{2} = C_0 e^{-t_{1/2}/\tau}.$$

Dividing by C_0 ,

$$\frac{1}{2} = e^{-t_{1/2}/\tau}.$$

Taking logarithms,

$$-\ln 2 = -\frac{t_{1/2}}{\tau},$$

so

$$t_{1/2} = \tau \ln 2.$$

For $\tau = 20$ ms,

$$t_{1/2} = 20 \ln 2 \approx 13.9 \text{ ms}.$$

Practice Questions

Question 2.1: Solving an exponential growth equation

Solve the initial-value problem

$$\frac{dy}{dt} = 3y, \quad y(0) = 2.$$

Verify your solution by differentiating it.

Question 2.2: Exponential decay and threshold time

A quantity $x(t)$ satisfies

$$\frac{dx}{dt} = -0.4x, \quad x(0) = 10.$$

- (a) Find $x(t)$.
- (b) Calculate $x(3)$.
- (c) Find the time at which $x(t) = 2$.

Question 2.3: Determining an unknown growth rate

A population satisfies

$$\frac{dN}{dt} = aN, \quad N(0) = 50.$$

After four time units, the population is 100.

- (a) Find the value of a .
- (b) Write the complete expression for $N(t)$.
- (c) Calculate $N(10)$.

Question 2.4: A time-dependent rate

Solve the initial-value problem

$$\frac{dy}{dt} = -2ty, \quad y(0) = 3.$$

Verify that your answer satisfies both the differential equation and the initial condition.

Question 2.5: Nonlinear clearance

Suppose a chemical concentration $C(t)$ is removed according to

$$\frac{dC}{dt} = -kC^2, \quad C(0) = C_0,$$

where $k > 0$.

- (a) Separate the variables and show that

$$C(t) = \frac{C_0}{1 + kC_0t}.$$

- (b) Find the time at which $C(t) = C_0/2$.

- (c) Let

$$C_0 = 4 \mu\text{M}, \quad k = 0.05 \mu\text{M}^{-1}\text{ms}^{-1}.$$

Calculate $C(10 \text{ ms})$.

- (d) Compare the long-term form of this decay with exponential decay.

2.2 First-Order Linear ODEs with Forcing

A first-order linear differential equation can be written in the form

$$\frac{dy}{dt} + p(t)y = q(t).$$

An integrating factor is

$$\mu(t) = e^{\int p(t) dt}.$$

Multiplying the differential equation by $\mu(t)$ allows the left-hand side to be written as

$$\frac{d}{dt}(\mu(t)y(t)).$$

Worked Example 2.3: Constant forcing and approach to equilibrium

Solve the initial-value problem

$$\frac{dy}{dt} + 2y = 6, \quad y(0) = 1.$$

Identify the steady-state value and the transient part of the solution.

Solution. The integrating factor is

$$\mu(t) = e^{\int 2 dt} = e^{2t}.$$

Multiplying the differential equation by e^{2t} gives

$$e^{2t} \frac{dy}{dt} + 2e^{2t}y = 6e^{2t}.$$

The left-hand side is the derivative of $e^{2t}y$:

$$\frac{d}{dt}(e^{2t}y) = 6e^{2t}.$$

Integrating,

$$e^{2t}y = \int 6e^{2t} dt = 3e^{2t} + C.$$

Dividing by e^{2t} ,

$$y(t) = 3 + Ce^{-2t}.$$

Using $y(0) = 1$,

$$1 = 3 + C,$$

so

$$C = -2.$$

Therefore,

$$\boxed{y(t) = 3 - 2e^{-2t}}.$$

The steady-state value is

$$y_{\infty} = 3.$$

The transient part is

$$-2e^{-2t},$$

which approaches zero as t increases. Thus the solution approaches the steady state exponentially.

Worked Example 2.4: Filtering a synaptic input

A synaptic variable $s(t)$ satisfies

$$\frac{ds}{dt} + \frac{1}{5}s = 0.2e^{-t/10}, \quad s(0) = 0,$$

where time is measured in milliseconds.

- (a) Find $s(t)$.
- (b) Find the time at which $s(t)$ reaches its maximum.
- (c) Calculate the maximum value.

Solution. The integrating factor is

$$\mu(t) = e^{\int (1/5) dt} = e^{t/5}.$$

Multiplying the equation by $e^{t/5}$ gives

$$e^{t/5} \frac{ds}{dt} + \frac{1}{5} e^{t/5} s = 0.2 e^{t/5} e^{-t/10}.$$

Therefore,

$$\frac{d}{dt} (e^{t/5} s) = 0.2 e^{t/10}.$$

Integrating,

$$e^{t/5} s = 2e^{t/10} + C.$$

Hence

$$s(t) = 2e^{-t/10} + Ce^{-t/5}.$$

Using $s(0) = 0$,

$$0 = 2 + C,$$

so

$$C = -2.$$

Thus

$$s(t) = 2 \left(e^{-t/10} - e^{-t/5} \right).$$

To find the maximum, differentiate:

$$\frac{ds}{dt} = -\frac{1}{5} e^{-t/10} + \frac{2}{5} e^{-t/5}.$$

At the maximum,

$$\frac{ds}{dt} = 0.$$

Therefore,

$$2e^{-t/5} = e^{-t/10}.$$

Dividing by $e^{-t/5}$,

$$2 = e^{t/10}.$$

Thus

$$t_{\max} = 10 \ln 2 \text{ ms}$$

or approximately

$$t_{\max} \approx 6.93 \text{ ms.}$$

At this time,

$$e^{-t_{\max}/10} = \frac{1}{2}, \quad e^{-t_{\max}/5} = \frac{1}{4}.$$

Therefore,

$$s(t_{\max}) = 2 \left(\frac{1}{2} - \frac{1}{4} \right) = \boxed{\frac{1}{2}}.$$

Practice Questions

Question 2.6: Constant forcing

Solve

$$\frac{dy}{dt} + 3y = 12, \quad y(0) = 2.$$

- Find $y(t)$.
- Identify the steady-state value.
- Find the time at which $y(t) = 3.5$.
- Verify the solution by substitution into the differential equation.

Question 2.7: Polynomial forcing

Solve the initial-value problem

$$\frac{dy}{dt} + y = t, \quad y(0) = 1.$$

Separate your answer into a forced part and a transient part, and describe the long-term behaviour.

Question 2.8: General constant-coefficient equation

Let $a > 0$, and consider

$$\frac{dy}{dt} + ay = b, \quad y(0) = y_0,$$

where a , b and y_0 are constants.

- Use an integrating factor to show that

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a} \right) e^{-at}.$$

- Identify the steady-state value.
- Identify the time constant.

- (d) Find the time at which the magnitude of the initial deviation from equilibrium has been reduced by one half.

Question 2.9: Exponentially decaying forcing

Solve

$$\frac{dy}{dt} + 2y = 4e^{-t}, \quad y(0) = 0.$$

- (a) Find $y(t)$.
- (b) Show that $y(t)$ initially increases but eventually approaches zero.
- (c) Find the time at which $y(t)$ reaches its maximum.
- (d) Calculate the maximum value.

Question 2.10: Adaptation of a firing rate

Following a sudden change in input, a firing rate $r(t)$ approaches a new steady value according to

$$\frac{dr}{dt} = \frac{r_\infty - r}{\tau}.$$

Let

$$r_\infty = 15 \text{ Hz}, \quad r(0) = 5 \text{ Hz}, \quad \tau = 20 \text{ ms}.$$

- (a) Rewrite the equation in standard linear form.
- (b) Find $r(t)$.
- (c) Calculate $r(20 \text{ ms})$.
- (d) Find the time at which the firing rate has completed 95% of its change from 5 Hz to 15 Hz.
- (e) Explain the meaning of the time constant τ in this model.

2.3 The Deterministic Leaky Integrate-and-Fire Model

The leaky integrate-and-fire model is

$$C_m \frac{dV}{dt} = -g_L(V - E_L) + I(t),$$

where $V(t)$ is the membrane potential, C_m is the membrane capacitance, g_L is the leak conductance, E_L is the leak reversal potential, and $I(t)$ is the applied current.

It is useful to define

$$\tau_m = \frac{C_m}{g_L}, \quad R_m = \frac{1}{g_L}.$$

For a constant input $I(t) = I_0$, the equation becomes

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I_0.$$

When $V(t)$ reaches a threshold V_{th} , the neuron emits a spike and the voltage is reset to V_{reset} .

Worked Example 2.5: Solving the subthreshold LIF equation

Consider

$$C_m \frac{dV}{dt} = -g_L(V - E_L) + I_0, \quad V(0) = V_0,$$

where I_0 is constant.

Derive the membrane-potential trajectory and identify its equilibrium value.

Solution. Divide the equation by g_L :

$$\frac{C_m}{g_L} \frac{dV}{dt} = -(V - E_L) + \frac{I_0}{g_L}.$$

Using

$$\tau_m = \frac{C_m}{g_L}, \quad R_m = \frac{1}{g_L},$$

gives

$$\tau_m \frac{dV}{dt} = -V + E_L + R_m I_0.$$

Define the equilibrium potential

$$V_\infty = E_L + R_m I_0.$$

The equation can then be written as

$$\frac{dV}{dt} = -\frac{1}{\tau_m}(V - V_\infty).$$

The solution is

$$V(t) = V_\infty + C e^{-t/\tau_m}.$$

Using $V(0) = V_0$,

$$V_0 = V_\infty + C,$$

so

$$C = V_0 - V_\infty.$$

Therefore,

$$V(t) = V_\infty + (V_0 - V_\infty) e^{-t/\tau_m}.$$

Since

$$e^{-t/\tau_m} \rightarrow 0$$

as $t \rightarrow \infty$,

$$V(t) \rightarrow V_\infty = E_L + R_m I_0.$$

The membrane time constant τ_m controls how quickly the voltage approaches this equilibrium.

Worked Example 2.6: Calculating the time to spike

A neuron has

$$E_L = -65 \text{ mV}, \quad \tau_m = 20 \text{ ms}, \quad R_m I_0 = 20 \text{ mV}.$$

Initially,

$$V_0 = -65 \text{ mV},$$

and the spike threshold is

$$V_{\text{th}} = -50 \text{ mV}.$$

Determine whether the neuron spikes and, if so, calculate the time to threshold.

Solution. The equilibrium potential is

$$V_{\infty} = E_L + R_m I_0 = -65 + 20 = -45 \text{ mV}.$$

Because

$$V_{\infty} > V_{\text{th}},$$

the voltage can reach threshold.

The voltage trajectory is

$$V(t) = -45 + (-65 - (-45)) e^{-t/20}.$$

Thus

$$V(t) = -45 - 20e^{-t/20}.$$

At threshold,

$$-50 = -45 - 20e^{-t_{\text{sp}}/20}.$$

Therefore,

$$-5 = -20e^{-t_{\text{sp}}/20},$$

so

$$e^{-t_{\text{sp}}/20} = \frac{1}{4}.$$

Taking logarithms,

$$-\frac{t_{\text{sp}}}{20} = \ln\left(\frac{1}{4}\right).$$

Hence

$$t_{\text{sp}} = -20 \ln\left(\frac{1}{4}\right) = 20 \ln 4.$$

Therefore,

$$\boxed{t_{\text{sp}} \approx 27.7 \text{ ms}}.$$

More generally, the time to threshold is

$$\boxed{t_{\text{sp}} = \tau_m \ln\left(\frac{V_{\infty} - V_0}{V_{\infty} - V_{\text{th}}}\right)},$$

provided that

$$V_0 < V_{\text{th}} < V_{\infty}.$$

Practice Questions

Question 2.11: Subthreshold response

A neuron satisfies

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I_0,$$

with

$$E_L = -70 \text{ mV}, \quad \tau_m = 10 \text{ ms}, \quad R_m I_0 = 5 \text{ mV}, \quad V(0) = -60 \text{ mV}.$$

- Find the equilibrium potential V_∞ .
- Find $V(t)$.
- Calculate $V(10 \text{ ms})$.
- Describe the long-term behaviour.

Question 2.12: Minimum current required for spiking

A neuron begins below threshold and receives a constant current. Its equilibrium potential is

$$V_\infty = E_L + R_m I_0.$$

- Explain why the neuron can reach threshold only if

$$V_\infty > V_{\text{th}}.$$

- Show that the condition for spiking is

$$R_m I_0 > V_{\text{th}} - E_L.$$

- Let

$$E_L = -65 \text{ mV}, \quad V_{\text{th}} = -50 \text{ mV}.$$

Find the minimum value of $R_m I_0$ required for threshold crossing.

- Explain what happens when

$$R_m I_0 = V_{\text{th}} - E_L.$$

Question 2.13: Time to threshold

A neuron has

$$E_L = -70 \text{ mV}, \quad \tau_m = 15 \text{ ms}, \quad R_m I_0 = 30 \text{ mV}.$$

The initial voltage and threshold are

$$V_0 = -60 \text{ mV}, \quad V_{\text{th}} = -50 \text{ mV}.$$

- Find V_∞ .
- Write down $V(t)$.

- (c) Calculate the time at which the voltage reaches threshold.
- (d) Verify that the threshold time is positive and finite.

Question 2.14: Reset, refractory period and firing rate

A neuron has

$$E_L = -65 \text{ mV}, \quad \tau_m = 20 \text{ ms}, \quad R_m I_0 = 25 \text{ mV}.$$

After each spike, it is reset to

$$V_{\text{reset}} = -65 \text{ mV}.$$

The threshold and absolute refractory period are

$$V_{\text{th}} = -50 \text{ mV}, \quad t_{\text{ref}} = 2 \text{ ms}.$$

Assume the constant input remains present after every reset.

- (a) Calculate V_{∞} .
- (b) Calculate the time from reset to threshold.
- (c) Find the total interspike interval, including the refractory period.
- (d) Calculate the firing rate in hertz.

Question 2.15: Effects of capacitance and leak conductance

Three neurons obey

$$C_m \frac{dV}{dt} = -g_L(V - E_L) + I_0.$$

For all three neurons,

$$E_L = -65 \text{ mV}, \quad V_0 = -65 \text{ mV}, \quad V_{\text{th}} = -50 \text{ mV}, \quad I_0 = 200 \text{ pA}.$$

Their parameters are:

	C_m	g_L
Neuron A	200 pF	10 nS
Neuron B	100 pF	10 nS
Neuron C	200 pF	20 nS

Use

$$\tau_m = \frac{C_m}{g_L}, \quad R_m I_0 = \frac{I_0}{g_L}.$$

- (a) Calculate τ_m , $R_m I_0$, and V_{∞} for each neuron.
- (b) Determine which neurons reach threshold.
- (c) For each neuron that spikes, calculate the time to threshold.
- (d) Explain the effect of reducing capacitance while holding conductance fixed.
- (e) Explain why increasing the leak conductance can prevent spiking even though it reduces the time constant.

2.4 Second-Order Linear ODEs and Oscillations

A second-order differential equation requires two initial conditions, usually the initial value and the initial derivative.

For an equation of the form

$$\frac{d^2y}{dt^2} = ay,$$

the behaviour depends on the sign of a .

If $a > 0$, the solutions are combinations of growing and decaying exponentials. If $a < 0$, the solutions are oscillatory and can be written using sine and cosine functions.

Worked Example 2.7: A second-order equation with exponential solutions

Solve

$$\frac{d^2y}{dt^2} = 4y,$$

subject to

$$y(0) = 3, \quad \frac{dy}{dt}(0) = -2.$$

Describe the long-term behaviour of the solution.

Solution. Try a solution of the form

$$y(t) = e^{rt}.$$

Substitution gives

$$r^2 e^{rt} = 4e^{rt},$$

so the characteristic equation is

$$r^2 = 4.$$

Therefore,

$$r = 2 \quad \text{or} \quad r = -2.$$

The general solution is

$$y(t) = Ae^{2t} + Be^{-2t}.$$

Differentiating,

$$\frac{dy}{dt} = 2Ae^{2t} - 2Be^{-2t}.$$

The initial condition $y(0) = 3$ gives

$$A + B = 3.$$

The initial derivative gives

$$2A - 2B = -2,$$

or

$$A - B = -1.$$

Solving these two equations,

$$A = 1, \quad B = 2.$$

Therefore,

$$\boxed{y(t) = e^{2t} + 2e^{-2t}}.$$

For large positive t , the term e^{2t} dominates. Hence the solution eventually grows exponentially.

Worked Example 2.8: an oscillating population mode

Let $x(t)$ represent the deviation of a neural population's activity from its baseline value. Suppose

$$\frac{d^2x}{dt^2} + 100\pi^2x = 0,$$

where t is measured in seconds, with

$$x(0) = 2, \quad \frac{dx}{dt}(0) = 10\pi.$$

Find $x(t)$, the angular frequency, the ordinary frequency, the period and the amplitude of the oscillation.

Solution. The characteristic equation is

$$r^2 + 100\pi^2 = 0.$$

Therefore,

$$r = \pm 10\pi i.$$

Using Euler's formula, the real-valued general solution can be written as

$$x(t) = A \cos(10\pi t) + B \sin(10\pi t).$$

The first initial condition gives

$$x(0) = A = 2.$$

Differentiating,

$$\frac{dx}{dt} = -10\pi A \sin(10\pi t) + 10\pi B \cos(10\pi t).$$

Thus

$$\frac{dx}{dt}(0) = 10\pi B.$$

Using the initial derivative,

$$10\pi B = 10\pi,$$

so

$$B = 1.$$

Therefore,

$$x(t) = 2 \cos(10\pi t) + \sin(10\pi t).$$

The angular frequency is

$$\omega = 10\pi \text{ rad s}^{-1}.$$

The ordinary frequency is

$$f = \frac{\omega}{2\pi} = \frac{10\pi}{2\pi} = 5 \text{ Hz}.$$

The period is

$$T = \frac{1}{f} = 0.2 \text{ s}.$$

A function of the form

$$A \cos(\omega t) + B \sin(\omega t)$$

has amplitude

$$\sqrt{A^2 + B^2}.$$

Therefore, the amplitude is

$$\sqrt{5}.$$

Practice Questions**Question 2.16: Exponential modes**

Solve

$$\frac{d^2y}{dt^2} = 9y,$$

subject to

$$y(0) = 2, \quad \frac{dy}{dt}(0) = -3.$$

- (a) Find the characteristic roots.
- (b) Find the general solution.
- (c) Apply the two initial conditions.
- (d) Describe which term dominates for large positive t .

Question 2.17: An oscillatory initial-value problem

Solve

$$\frac{d^2y}{dt^2} + 16y = 0,$$

subject to

$$y(0) = -1, \quad \frac{dy}{dt}(0) = 8.$$

- (a) Find $y(t)$.
- (b) Find the angular frequency.
- (c) Find the period.
- (d) Find the amplitude of the oscillation.

Question 2.18: Classifying behaviour from a parameter

Consider

$$\frac{d^2y}{dt^2} = ay.$$

For each of the following values of a , write down the general form of the solution and describe its qualitative behaviour:

- (a) $a = 4$,
- (b) $a = 0$,
- (c) $a = -9$.

Explain why oscillations occur only in one of the three cases.

Question 2.19: Euler's formula and oscillatory solutions

Euler's formula is

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Let

$$z(t) = e^{i\omega t},$$

where ω is real.

(a) Calculate $z'(t)$ and $z''(t)$.

(b) Show that

$$z''(t) = -\omega^2 z(t).$$

(c) Use Euler's formula to show that both

$$\cos(\omega t) \quad \text{and} \quad \sin(\omega t)$$

solve

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0.$$

Question 2.20: A gamma-frequency rhythm

A simplified neural population rhythm is modelled by

$$\frac{d^2 x}{dt^2} + (80\pi)^2 x = 0,$$

where t is measured in seconds.

Initially,

$$x(0) = 1, \quad \frac{dx}{dt}(0) = 0.$$

(a) Find $x(t)$.

(b) Find the angular frequency.

(c) Show that the ordinary frequency is 40 Hz.

(d) Find the period in milliseconds.

(e) Find the first positive time at which $x(t) = 0$.

(f) Find the first positive time at which $x(t) = -1$.

2.5 Euler's Method

For an initial-value problem

$$\frac{dy}{dt} = f(y, t), \quad y(t_0) = y_0,$$

Euler's method uses the update

$$y_{n+1} = y_n + hf(y_n, t_n),$$

where

$$t_{n+1} = t_n + h$$

and h is the time-step.

Worked Example 2.9: Euler approximation of exponential decay

Consider

$$\frac{dy}{dt} = -2y, \quad y(0) = 1.$$

Use Euler's method with step size

$$h = 0.25$$

to approximate $y(1)$. Compare the approximation with the exact solution.

Solution. Here,

$$f(y, t) = -2y,$$

so the Euler update is

$$y_{n+1} = y_n + 0.25(-2y_n).$$

Therefore,

$$y_{n+1} = 0.5y_n.$$

Starting from

$$t_0 = 0, \quad y_0 = 1,$$

the first step gives

$$y_1 = 0.5(1) = 0.5.$$

At the next step,

$$y_2 = 0.5(0.5) = 0.25.$$

Continuing,

$$y_3 = 0.125, \quad y_4 = 0.0625.$$

The values can be summarised as

t_n	y_n
0	1
0.25	0.5
0.50	0.25
0.75	0.125
1.00	0.0625

Thus Euler's method gives

$$y(1) \approx 0.0625.$$

The exact solution is

$$y(t) = e^{-2t},$$

so

$$y(1) = e^{-2} \approx 0.1353.$$

The absolute error is therefore

$$|0.0625 - 0.1353| \approx 0.0728.$$

Euler's method follows the decay qualitatively, but the chosen step size causes it to underestimate the solution.

Worked Example 2.10: numerically evolving an LIF voltage

Consider the subthreshold leaky integrate-and-fire equation

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I_0,$$

with

$$E_L = -65 \text{ mV}, \quad R_m I_0 = 20 \text{ mV}, \quad \tau_m = 10 \text{ ms}.$$

The initial voltage is

$$V(0) = -65 \text{ mV}.$$

Use Euler's method with

$$h = 2 \text{ ms}$$

to approximate the voltage at $t = 10 \text{ ms}$. Compare the result with the exact solution.

Solution. The equilibrium voltage is

$$V_\infty = E_L + R_m I_0 = -45 \text{ mV}.$$

The differential equation can be written as

$$\frac{dV}{dt} = \frac{V_\infty - V}{\tau_m} = \frac{-45 - V}{10}.$$

The Euler update is therefore

$$V_{n+1} = V_n + 2 \left(\frac{-45 - V_n}{10} \right).$$

Starting from

$$V_0 = -65,$$

we obtain

$$V_1 = -65 + 2 \left(\frac{20}{10} \right) = -61.$$

The next update is

$$V_2 = -61 + 2 \left(\frac{16}{10} \right) = -57.8.$$

Continuing,

$$V_3 = -57.8 + 2 \left(\frac{12.8}{10} \right) = -55.24,$$

$$V_4 = -55.24 + 2 \left(\frac{10.24}{10} \right) = -53.192,$$

and

$$V_5 = -53.192 + 2 \left(\frac{8.192}{10} \right) = -51.5536.$$

Thus

$$\boxed{V(10 \text{ ms}) \approx -51.55 \text{ mV}}.$$

The exact voltage is

$$V(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau_m}.$$

Therefore,

$$V(10) = -45 - 20e^{-1} \approx -52.36 \text{ mV}.$$

The Euler approximation differs from the exact value by approximately

$$0.80 \text{ mV}.$$

Practice Questions**Question 2.21: Carrying out Euler steps**

Consider

$$\frac{dy}{dt} = -y, \quad y(0) = 2.$$

Use Euler's method with

$$h = 0.5$$

to approximate the solution at

$$t = 0.5, \quad 1, \quad 1.5, \quad 2.$$

Compare your approximation at $t = 2$ with the exact value

$$y(2) = 2e^{-2}.$$

Question 2.22: Comparing two step sizes

Consider

$$\frac{dy}{dt} = -2y, \quad y(0) = 1.$$

- (a) Use Euler's method with $h = 0.5$ to approximate $y(1)$.
- (b) Repeat using $h = 0.25$.
- (c) Calculate the exact value $y(1)$.
- (d) Calculate the absolute error for each step size.
- (e) Which approximation is more accurate?

Question 2.23: A time-dependent differential equation

Consider

$$\frac{dy}{dt} = t - y, \quad y(0) = 1.$$

Use Euler's method with

$$h = 0.25$$

to approximate $y(1)$.

The exact solution is

$$y(t) = t - 1 + 2e^{-t}.$$

Compare the numerical and exact values at $t = 1$.

Question 2.24: Step size and numerical stability

Consider

$$\frac{dy}{dt} = -4y, \quad y(0) = 1.$$

- (a) Show that the Euler update is

$$y_{n+1} = (1 - 4h)y_n.$$

- (b) Describe the numerical behaviour when $h = 0.2$.
 (c) Describe the numerical behaviour when $h = 0.4$.
 (d) Describe the numerical behaviour when $h = 0.6$.
 (e) Show that the Euler approximation decays to zero only when

$$|1 - 4h| < 1.$$

- (f) Find the range of positive step sizes satisfying this condition.

Question 2.25: Euler simulation of threshold crossing

A leaky integrate-and-fire neuron satisfies

$$\frac{dV}{dt} = \frac{V_\infty - V}{\tau_m},$$

where

$$V_\infty = -40 \text{ mV}, \quad \tau_m = 20 \text{ ms}.$$

The initial voltage and threshold are

$$V(0) = -70 \text{ mV}, \quad V_{\text{th}} = -50 \text{ mV}.$$

- (a) Write down the Euler update for a step size of

$$h = 5 \text{ ms}.$$

- (b) Calculate the numerical voltage at

$$t = 5, \quad 10, \quad 15, \quad 20 \text{ ms}.$$

- (c) According to the Euler approximation, when is threshold first crossed?
 (d) Show that the exact threshold time is

$$t_{\text{sp}} = 20 \ln 3 \text{ ms}.$$

- (e) Compare the Euler estimate with the exact threshold time.
 (f) Explain why a smaller step size would give a more reliable estimate of the spike time.

2.6 Runge–Kutta Methods and Numerical Accuracy

For

$$\frac{dy}{dt} = f(y, t), \quad y(t_n) = y_n,$$

the classical fourth-order Runge–Kutta method, usually called RK4, uses

$$\begin{aligned} k_1 &= hf(y_n, t_n), \\ k_2 &= hf\left(y_n + \frac{k_1}{2}, t_n + \frac{h}{2}\right), \\ k_3 &= hf\left(y_n + \frac{k_2}{2}, t_n + \frac{h}{2}\right), \\ k_4 &= hf(y_n + k_3, t_n + h), \end{aligned}$$

followed by the update

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

In this convention, the quantities $k_1, \dots, k_4$ already include the step size h .

For sufficiently smooth problems, RK4 has global error of order

$$O(h^4).$$

Worked Example 2.11: One RK4 step for exponential decay

Consider

$$\frac{dy}{dt} = -y, \quad y(0) = 1.$$

Use one RK4 step of size

$$h = 0.5$$

to approximate $y(0.5)$. Compare the result with the exact solution.

Solution. Here,

$$f(y, t) = -y.$$

The first increment is

$$k_1 = 0.5(-1) = -0.5.$$

For the second increment,

$$y_0 + \frac{k_1}{2} = 1 - \frac{0.5}{2} = 0.75.$$

Therefore,

$$k_2 = 0.5(-0.75) = -0.375.$$

For the third increment,

$$y_0 + \frac{k_2}{2} = 1 - \frac{0.375}{2} = 0.8125,$$

so

$$k_3 = 0.5(-0.8125) = -0.40625.$$

For the fourth increment,

$$y_0 + k_3 = 1 - 0.40625 = 0.59375,$$

and hence

$$k_4 = 0.5(-0.59375) = -0.296875.$$

The RK4 update is

$$y_1 = 1 + \frac{1}{6}(-0.5 + 2(-0.375) + 2(-0.40625) - 0.296875).$$

Thus

$$y(0.5) \approx 0.6067708.$$

The exact solution is

$$y(t) = e^{-t},$$

so

$$y(0.5) = e^{-0.5} \approx 0.6065307.$$

The absolute error is approximately

$$2.40 \times 10^{-4}.$$

Worked Example 2.12: an RK4 step for an LIF voltage

Consider the subthreshold equation

$$\frac{dV}{dt} = \frac{V_\infty - V}{\tau_m},$$

where

$$V_\infty = -45 \text{ mV}, \quad \tau_m = 10 \text{ ms}.$$

The initial voltage is

$$V(0) = -65 \text{ mV}.$$

Use one RK4 step of size

$$h = 5 \text{ ms}$$

to approximate $V(5 \text{ ms})$. Compare the result with Euler's method and the exact solution.

Solution. Here,

$$f(V, t) = \frac{-45 - V}{10}.$$

The first increment is

$$k_1 = 5 \left(\frac{-45 - (-65)}{10} \right) = 10.$$

For the second increment,

$$V_0 + \frac{k_1}{2} = -65 + 5 = -60.$$

Thus

$$k_2 = 5 \left(\frac{-45 - (-60)}{10} \right) = 7.5.$$

For the third increment,

$$V_0 + \frac{k_2}{2} = -65 + 3.75 = -61.25,$$

so

$$k_3 = 5 \left(\frac{-45 - (-61.25)}{10} \right) = 8.125.$$

For the fourth increment,

$$V_0 + k_3 = -65 + 8.125 = -56.875.$$

Therefore,

$$k_4 = 5 \left(\frac{-45 - (-56.875)}{10} \right) = 5.9375.$$

The RK4 update is

$$V_1 = -65 + \frac{1}{6} (10 + 2(7.5) + 2(8.125) + 5.9375).$$

Hence

$$V(5 \text{ ms}) \approx -57.1354 \text{ mV}.$$

The exact solution is

$$V(t) = -45 - 20e^{-t/10}.$$

Therefore,

$$V(5) = -45 - 20e^{-0.5} \approx -57.1306 \text{ mV}.$$

The RK4 absolute error is approximately

$$0.0048 \text{ mV}.$$

A single Euler step gives

$$V_{\text{Euler}} = -65 + 5 \left(\frac{20}{10} \right) = -55 \text{ mV}.$$

Its absolute error is approximately

$$2.13 \text{ mV}.$$

Practice Questions

Question 2.26: One RK4 step with time-dependent forcing

Consider

$$\frac{dy}{dt} = t + y, \quad y(0) = 1.$$

Use one RK4 step with

$$h = 0.2$$

to approximate $y(0.2)$.

The exact solution is

$$y(t) = 2e^t - t - 1.$$

Compare the RK4 approximation with the exact value.

Question 2.27: Two RK4 steps and comparison with Euler's method

Consider

$$\frac{dy}{dt} = -2y, \quad y(0) = 1.$$

- (a) Use RK4 with $h = 0.5$ to approximate $y(0.5)$.
- (b) Take a second RK4 step to approximate $y(1)$.
- (c) Calculate the exact value

$$y(1) = e^{-2}.$$
- (d) Calculate the RK4 absolute error.
- (e) Use Euler's method with the same step size to approximate $y(1)$.
- (f) Compare the two numerical methods.

Question 2.28: Completing an RK4 calculation

Consider

$$\frac{dy}{dt} = t - y, \quad y(0) = 1,$$

with step size

$$h = 0.2.$$

The first two RK4 increments are

$$k_1 = -0.2, \quad k_2 = -0.16.$$

- (a) Calculate k_3 .
- (b) Calculate k_4 .
- (c) Calculate the RK4 approximation to $y(0.2)$.
- (d) The exact solution is

$$y(t) = t - 1 + 2e^{-t}.$$

Calculate the absolute error.

Question 2.29: Error reduction when the step size is halved

For the problem

$$\frac{dy}{dt} = -y, \quad y(0) = 1,$$

the exact value at $t = 1$ is

$$y(1) = e^{-1} \approx 0.3678794412.$$

RK4 gives the following approximations:

h	RK4 approximation to $y(1)$
0.5	0.3681708442
0.25	0.3678941994
0.125	0.3678802719

- (a) Calculate the absolute error for each step size.
- (b) Calculate
- $$\frac{E_{0.5}}{E_{0.25}} \quad \text{and} \quad \frac{E_{0.25}}{E_{0.125}},$$
- where E_h is the error obtained with step size h .
- (c) For a fourth-order method, by what factor should the error approximately decrease when h is halved?
- (d) Are the numerical results reasonably consistent with fourth-order convergence?

Question 2.30: Filtering a synaptic input

A synaptic variable satisfies

$$\frac{ds}{dt} = -\frac{s}{5} + 0.2e^{-t/10}, \quad s(0) = 0,$$

where t is measured in milliseconds.

- (a) Use one RK4 step with
- $$h = 2 \text{ ms}$$
- to approximate $s(2 \text{ ms})$.
- (b) The exact solution is
- $$s(t) = 2 \left(e^{-t/10} - e^{-t/5} \right).$$
- Calculate the exact value at $t = 2 \text{ ms}$.
- (c) Calculate the RK4 absolute error.
- (d) Use one Euler step with the same step size to approximate $s(2 \text{ ms})$.
- (e) Compare the Euler and RK4 errors.

2.7 Linear Systems of ODEs

A system of two linear differential equations can be written as

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}, \quad \mathbf{u}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}.$$

If A has two linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, with eigenvalues λ_1, λ_2 , then the general solution is

$$\mathbf{u}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2.$$

The constants c_1 and c_2 are determined by the initial condition.

Worked Example 2.13: Solving a coupled linear system

Consider

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= -2x - 3y,\end{aligned}$$

subject to

$$x(0) = 2, \quad y(0) = -1.$$

Solve the system and describe its long-term behaviour.

Solution. Define

$$\mathbf{u}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}.$$

The system can be written as

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}, \quad A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}.$$

The eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

Thus

$$\det \begin{pmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{pmatrix} = 0.$$

Therefore,

$$\lambda(3 + \lambda) + 2 = 0,$$

so

$$\lambda^2 + 3\lambda + 2 = 0.$$

Factorising,

$$(\lambda + 1)(\lambda + 2) = 0.$$

Hence

$$\lambda_1 = -1, \quad \lambda_2 = -2.$$

For $\lambda_1 = -1$,

$$(A + I)\mathbf{v}_1 = \mathbf{0}.$$

This gives

$$x + y = 0,$$

so one eigenvector is

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

For $\lambda_2 = -2$,

$$(A + 2I)\mathbf{v}_2 = \mathbf{0},$$

which gives

$$2x + y = 0.$$

One corresponding eigenvector is

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

The general solution is therefore

$$\mathbf{u}(t) = c_1 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

At $t = 0$,

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Comparing components,

$$c_1 + c_2 = 2,$$

and

$$-c_1 - 2c_2 = -1.$$

Solving gives

$$c_1 = 3, \quad c_2 = -1.$$

Therefore,

$$\mathbf{u}(t) = 3e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

The individual components are

$$x(t) = 3e^{-t} - e^{-2t}$$

and

$$y(t) = -3e^{-t} + 2e^{-2t}.$$

Both exponential terms approach zero. Hence

$$\mathbf{u}(t) \longrightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

as $t \longrightarrow \infty$.

The mode associated with $\lambda = -1$ decays more slowly and therefore dominates the long-term direction of approach.

Worked Example 2.14: coupled population modes

Let

$$\mathbf{r}(t) = \begin{pmatrix} r_1(t) \\ r_2(t) \end{pmatrix}$$

represent deviations of two neural population activities from their baseline values.

Suppose

$$\frac{d\mathbf{r}}{dt} = A\mathbf{r}, \quad A = \begin{pmatrix} -1 & 0.5 \\ 0.5 & -1 \end{pmatrix},$$

with

$$\mathbf{r}(0) = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

Find $\mathbf{r}(t)$ and interpret the two activity modes.

Solution. Consider the common-activity vector

$$\mathbf{v}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Multiplying,

$$A\mathbf{v}_+ = \begin{pmatrix} -0.5 \\ -0.5 \end{pmatrix} = -0.5\mathbf{v}_+.$$

Thus

$$\lambda_+ = -0.5.$$

Now consider the difference vector

$$\mathbf{v}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then

$$A\mathbf{v}_- = \begin{pmatrix} -1.5 \\ 1.5 \end{pmatrix} = -1.5\mathbf{v}_-.$$

Thus

$$\lambda_- = -1.5.$$

The general solution is

$$\mathbf{r}(t) = c_+ e^{-0.5t} \mathbf{v}_+ + c_- e^{-1.5t} \mathbf{v}_-.$$

Decompose the initial condition:

$$\begin{pmatrix} 4 \\ 2 \end{pmatrix} = c_+ \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_- \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus

$$c_+ + c_- = 4, \quad c_+ - c_- = 2.$$

Therefore,

$$c_+ = 3, \quad c_- = 1.$$

Hence

$$\mathbf{r}(t) = 3e^{-0.5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + e^{-1.5t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The common-activity mode decays with rate 0.5, while the difference between the two populations decays with rate 1.5.

The difference mode disappears more quickly. The two populations therefore become increasingly similar before both activities eventually return to baseline.

Practice Questions

Question 2.31: Writing a system in matrix form

Consider

$$\begin{aligned}\frac{dx}{dt} &= 4x - y, \\ \frac{dy}{dt} &= 2x + y.\end{aligned}$$

(a) Define

$$\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}$$

and write the system as

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}.$$

(b) Calculate the derivative vector when

$$\mathbf{u} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

(c) Interpret the two components of the derivative vector.

Question 2.32: Solving a system using eigenmodes

Consider

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}, \quad A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix},$$

with

$$\mathbf{u}(0) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(a) Verify that

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

are eigenvectors of A .

(b) Find the associated eigenvalues.

(c) Express the initial condition as a linear combination of the two eigenvectors.

(d) Find $\mathbf{u}(t)$.

(e) Which mode dominates for large positive t ?

Question 2.33: Converting a second-order equation into a system

Consider

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0,$$

with

$$x(0) = 1, \quad \frac{dx}{dt}(0) = 0.$$

Define

$$y = \frac{dx}{dt}.$$

- Write the second-order equation as a first-order system for x and y .
- Write the system in matrix form.
- Find the eigenvalues and eigenvectors of the matrix.
- Solve for $x(t)$ and $y(t)$.
- Verify that your expression for $x(t)$ satisfies the original second-order equation.

Question 2.34: Stable and unstable directions

Consider

$$\begin{aligned} \frac{dx}{dt} &= x, \\ \frac{dy}{dt} &= -2y, \end{aligned}$$

with

$$x(0) = -1, \quad y(0) = 3.$$

- Write the system in matrix form.
- Find $x(t)$ and $y(t)$.
- Identify the stable eigenvector direction.
- Identify the unstable eigenvector direction.
- Describe the long-term behaviour of the given solution.
- Explain why the origin is called a saddle point.

Question 2.35: Decay of common and difference modes

Two neural population deviations satisfy

$$\frac{d\mathbf{r}}{dt} = A\mathbf{r}, \quad A = \begin{pmatrix} -0.8 & 0.2 \\ 0.2 & -0.8 \end{pmatrix}.$$

The eigenpairs are

$$\lambda_+ = -0.6, \quad \mathbf{v}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and

$$\lambda_- = -1, \quad \mathbf{v}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The initial activity is

$$\mathbf{r}(0) = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

(a) Express the initial activity as a linear combination of $\mathbf{v}_+$ and $\mathbf{v}_-$.

(b) Find $\mathbf{r}(t)$.

(c) Find an expression for

$$r_1(t) + r_2(t).$$

(d) Find an expression for

$$r_1(t) - r_2(t).$$

(e) Find the time at which the difference between the population activities has fallen to 10% of its initial value.

(f) Describe the relative activity pattern for large t .

2.8 Equilibria, Jacobians, and Local Stability

Consider a nonlinear system

$$\begin{aligned}\frac{dx}{dt} &= f(x, y), \\ \frac{dy}{dt} &= g(x, y).\end{aligned}$$

An equilibrium (x^*, y^*) satisfies

$$f(x^*, y^*) = 0, \quad g(x^*, y^*) = 0.$$

The Jacobian matrix is

$$J(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}.$$

The eigenvalues of $J(x^*, y^*)$ describe the behaviour of small perturbations near the equilibrium.

Worked Example 2.15: Two equilibria with different stability

Consider

$$\begin{aligned}\frac{dx}{dt} &= x - x^2, \\ \frac{dy}{dt} &= -2y.\end{aligned}$$

Find all equilibria and classify their local stability.

Solution. At an equilibrium,

$$x - x^2 = 0$$

and

$$-2y = 0.$$

The first equation gives

$$x(1 - x) = 0,$$

so

$$x = 0 \quad \text{or} \quad x = 1.$$

The second equation gives

$$y = 0.$$

Therefore, the equilibria are

$$\boxed{(0, 0)} \quad \text{and} \quad \boxed{(1, 0)}.$$

The Jacobian is

$$J(x, y) = \begin{pmatrix} 1 - 2x & 0 \\ 0 & -2 \end{pmatrix}.$$

At $(0, 0)$,

$$J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$

The eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = -2.$$

One eigenvalue is positive and the other is negative. Therefore,

$$\boxed{(0, 0) \text{ is an unstable saddle point.}}$$

At $(1, 0)$,

$$J(1, 0) = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}.$$

Both eigenvalues are negative:

$$\lambda_1 = -1, \quad \lambda_2 = -2.$$

Therefore,

$$\boxed{(1, 0) \text{ is a locally asymptotically stable node.}}$$

Near $(1, 0)$, perturbations in both directions decay. The component associated with $\lambda = -1$ decays more slowly and dominates the long-term approach to equilibrium.

Worked Example 2.16: membrane and adaptation dynamics

Let $v(t)$ represent a small deviation of membrane potential from a baseline value, and let $w(t)$ represent an adaptation variable. Suppose

$$\begin{aligned} \frac{dv}{dt} &= -v - w, \\ \frac{dw}{dt} &= v - w^3. \end{aligned}$$

Find the equilibrium and determine its local stability.

Solution. At an equilibrium,

$$-v - w = 0$$

and

$$v - w^3 = 0.$$

The first equation gives

$$v = -w,$$

while the second gives

$$v = w^3.$$

Therefore,

$$-w = w^3.$$

Hence

$$w(w^2 + 1) = 0.$$

Since w is real,

$$w = 0.$$

It follows that

$$v = 0.$$

Thus the only equilibrium is

$$(v^*, w^*) = (0, 0).$$

The Jacobian is

$$J(v, w) = \begin{pmatrix} -1 & -1 \\ 1 & -3w^2 \end{pmatrix}.$$

At the equilibrium,

$$J(0, 0) = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic equation is

$$\det \begin{pmatrix} -1 - \lambda & -1 \\ 1 & -\lambda \end{pmatrix} = 0.$$

Therefore,

$$\lambda(1 + \lambda) + 1 = 0,$$

or

$$\lambda^2 + \lambda + 1 = 0.$$

The eigenvalues are

$$\lambda = \frac{-1 \pm \sqrt{1 - 4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}.$$

Both eigenvalues have negative real part. Therefore,

$$(0, 0) \text{ is a locally asymptotically stable spiral.}$$

Small perturbations produce damped oscillations in v and w before the system returns to equilibrium.

Practice Questions

Question 2.36: A stable linear equilibrium

Consider

$$\begin{aligned}\frac{dx}{dt} &= -x + 2y, \\ \frac{dy}{dt} &= -3y.\end{aligned}$$

- (a) Find the equilibrium.
- (b) Calculate the Jacobian.
- (c) Find the eigenvalues of the Jacobian.
- (d) Classify the equilibrium.
- (e) Which mode decays most slowly?

Question 2.37: A saddle and a stable node

Consider

$$\begin{aligned}\frac{dx}{dt} &= x(2 - x), \\ \frac{dy}{dt} &= -y.\end{aligned}$$

- (a) Find all equilibria.
- (b) Calculate the Jacobian matrix.
- (c) Evaluate the Jacobian at each equilibrium.
- (d) Find the corresponding eigenvalues.
- (e) Classify each equilibrium.

Question 2.38: Three equilibria in a nonlinear system

Consider

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= x - x^3 - y.\end{aligned}$$

- (a) Find all equilibria.
- (b) Show that the Jacobian is

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ 1 - 3x^2 & -1 \end{pmatrix}.$$

- (c) Find the eigenvalues at the equilibrium $(0, 0)$.
- (d) Find the eigenvalues at the equilibria $(1, 0)$ and $(-1, 0)$.
- (e) Classify all three equilibria.

Question 2.39: Classifying equilibria from eigenvalues

For each pair of eigenvalues below, classify the corresponding equilibrium and describe the behaviour of a small perturbation.

(a)

$$\lambda_1 = -2, \quad \lambda_2 = -5.$$

(b)

$$\lambda_1 = 1, \quad \lambda_2 = -4.$$

(c)

$$\lambda_{1,2} = 0.3 \pm 2i.$$

(d)

$$\lambda_{1,2} = \pm 3i.$$

Explain why the final case cannot, in general, be classified using linearisation alone.

Question 2.40: Excitatory activity and inhibitory feedback

Let $E(t)$ represent excitatory population activity and $I(t)$ represent inhibitory feedback. Consider

$$\begin{aligned} \frac{dE}{dt} &= E(2 - E) - I, \\ \frac{dI}{dt} &= E - I. \end{aligned}$$

(a) Find all equilibria.

(b) Show that the Jacobian is

$$J(E, I) = \begin{pmatrix} 2 - 2E & -1 \\ 1 & -1 \end{pmatrix}.$$

(c) Find and classify the equilibrium at $(0, 0)$.(d) Find and classify the equilibrium at $(1, 1)$.

(e) Describe the response to a small perturbation near each equilibrium.

2.9 Parameter Changes and Bifurcations

A dynamical system may depend on a parameter μ :

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}; \mu).$$

A bifurcation occurs when changing μ changes the number, stability or qualitative behaviour of the system's equilibria or long-term solutions.

For a scalar equation

$$\frac{dx}{dt} = f(x; \mu),$$

an equilibrium x^* is locally stable when

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} < 0,$$

and unstable when

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} > 0.$$

Worked Example 2.17: A saddle-node bifurcation

Consider

$$\begin{aligned} \frac{dx}{dt} &= \mu - x^2, \\ \frac{dy}{dt} &= -y. \end{aligned}$$

Find all equilibria as a function of μ , classify their stability, and identify the bifurcation.

Solution. At an equilibrium,

$$\mu - x^2 = 0$$

and

$$-y = 0.$$

Therefore,

$$x^2 = \mu, \quad y = 0.$$

The number of equilibria depends on the sign of μ .

If $\mu < 0$, there are no real solutions of $x^2 = \mu$, so there are no equilibria.

If $\mu = 0$, there is one equilibrium:

$$(0, 0).$$

If $\mu > 0$, there are two equilibria:

$$(\sqrt{\mu}, 0) \quad \text{and} \quad (-\sqrt{\mu}, 0).$$

The Jacobian is

$$J(x, y) = \begin{pmatrix} -2x & 0 \\ 0 & -1 \end{pmatrix}.$$

At

$$(\sqrt{\mu}, 0),$$

the eigenvalues are

$$\lambda_1 = -2\sqrt{\mu}, \quad \lambda_2 = -1.$$

Both eigenvalues are negative, so

$$(\sqrt{\mu}, 0) \text{ is a locally asymptotically stable node.}$$

At

$$(-\sqrt{\mu}, 0),$$

the eigenvalues are

$$\lambda_1 = 2\sqrt{\mu}, \quad \lambda_2 = -1.$$

The eigenvalues have opposite signs, so

$$(-\sqrt{\mu}, 0) \text{ is an unstable saddle point.}$$

At $\mu = 0$, one eigenvalue is zero:

$$\lambda_1 = 0, \quad \lambda_2 = -1.$$

Linearisation alone is inconclusive at this point. However, the complete equilibrium analysis shows that two equilibria are created as μ increases through zero.

Therefore,

$$\boxed{\mu = 0}$$

is a saddle-node bifurcation.

In a bifurcation diagram, the stable branch

$$x^* = \sqrt{\mu}$$

would be drawn as a solid curve, while the unstable branch

$$x^* = -\sqrt{\mu}$$

would be drawn as a dashed curve.

Worked Example 2.18: Supercritical Hopf bifurcation

Consider the nonlinear system

$$\begin{aligned}\frac{dx}{dt} &= \mu x - y - x(x^2 + y^2), \\ \frac{dy}{dt} &= x + \mu y - y(x^2 + y^2).\end{aligned}$$

Determine how the behaviour near the origin changes as μ passes through zero.

Solution. The origin is an equilibrium for every value of μ .

The Jacobian at the origin is

$$J(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}.$$

The characteristic equation is

$$(\mu - \lambda)^2 + 1 = 0,$$

so the eigenvalues are

$$\boxed{\lambda_{1,2} = \mu \pm i}.$$

If $\mu < 0$, the eigenvalues have negative real part. The origin is therefore a stable spiral.

If $\mu > 0$, the eigenvalues have positive real part. The origin is therefore an unstable spiral.

At $\mu = 0$, the eigenvalues are purely imaginary, so nonlinear terms must be considered.

Define the radial coordinate

$$r = \sqrt{x^2 + y^2}.$$

Since

$$r \frac{dr}{dt} = x \frac{dx}{dt} + y \frac{dy}{dt},$$

substitution gives

$$r \frac{dr}{dt} = \mu(x^2 + y^2) - (x^2 + y^2)^2.$$

Using $x^2 + y^2 = r^2$,

$$r \frac{dr}{dt} = \mu r^2 - r^4.$$

For $r > 0$,

$$\frac{dr}{dt} = \mu r - r^3 = r(\mu - r^2).$$

The angular coordinate satisfies

$$\frac{d\theta}{dt} = 1.$$

For $\mu < 0$, the only non-negative radial equilibrium is

$$r = 0,$$

and it is stable.

For $\mu > 0$, the radial equilibria are

$$r = 0 \quad \text{and} \quad r = \sqrt{\mu}.$$

The origin becomes unstable, while

$$r = \sqrt{\mu}$$

is stable. It represents a stable periodic orbit in the (x, y) -plane.

Therefore, a stable limit cycle is created as μ increases through zero. This is a

supercritical Hopf bifurcation.

Practice Questions

Question 2.41: A one-dimensional saddle-node bifurcation

Consider

$$\frac{dx}{dt} = \mu - 4x^2.$$

- Find all equilibria as a function of μ .
- Determine the number of equilibria when $\mu < 0$, $\mu = 0$ and $\mu > 0$.
- Calculate

$$\frac{\partial f}{\partial x}.$$

- Classify the stability of each equilibrium for $\mu > 0$.
- Identify the bifurcation value.
- Sketch the equilibrium branches against μ , using a solid line for stable equilibria and a dashed line for unstable equilibria.

Question 2.42: A transcritical bifurcation

Consider

$$\frac{dx}{dt} = \mu x - x^2.$$

- (a) Show that the equilibrium branches are

$$x_1^* = 0 \quad \text{and} \quad x_2^* = \mu.$$

- (b) Calculate

$$\frac{\partial f}{\partial x}.$$

- (c) Determine the stability of both branches when $\mu < 0$.
 (d) Determine the stability of both branches when $\mu > 0$.
 (e) Describe what happens at $\mu = 0$.
 (f) Sketch the bifurcation diagram.

Question 2.43: A supercritical pitchfork bifurcation

Consider

$$\frac{dx}{dt} = \mu x - x^3.$$

- (a) Find all equilibria when $\mu < 0$.
 (b) Find all equilibria when $\mu > 0$.
 (c) Calculate
- $$\frac{\partial f}{\partial x}.$$
- (d) Classify the stability of every equilibrium.
 (e) Explain why $\mu = 0$ is called a supercritical pitchfork bifurcation.
 (f) Sketch the bifurcation diagram.

Question 2.44: Oscillatory stability change

Consider the parameter-dependent linear system

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \mu & -2 \\ 2 & \mu \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

- (a) Show that the origin is the only equilibrium.
 (b) Find the eigenvalues.
 (c) Classify the origin when $\mu < 0$, $\mu = 0$ and $\mu > 0$.
 (d) Show that the radial distance satisfies

$$\frac{dr}{dt} = \mu r.$$

- (e) Show that the angular velocity is

$$\frac{d\theta}{dt} = 2.$$

- (f) Explain why this linear system does not produce a stable finite-amplitude oscillation when $\mu > 0$.
- (g) What kind of nonlinear term could prevent the amplitude from growing without bound?

Question 2.45: Onset of a population rhythm

Let $A(t) \geq 0$ represent the amplitude of a neural population oscillation. Suppose

$$\frac{dA}{dt} = (I - I_c)A - A^3,$$

where I is a dimensionless input drive and I_c is its critical value.

- (a) Find all non-negative equilibrium amplitudes when $I < I_c$.
- (b) Find all non-negative equilibrium amplitudes when $I > I_c$.
- (c) Determine the stability of $A = 0$ on either side of I_c .
- (d) Determine the stability of the non-zero equilibrium amplitude.
- (e) Show that, above the critical input,

$$A^* = \sqrt{I - I_c}.$$

- (f) Calculate the equilibrium amplitude when

$$I - I_c = 9.$$

- (g) Interpret the bifurcation in terms of the onset of a neural rhythm.

3 Stochastic ODEs

3.1 Comparing ODEs and SDEs

A deterministic ordinary differential equation can be written as

$$dy = f(y, t) dt.$$

For a small time-step h , its Euler update is

$$y_{n+1} = y_n + f(y_n, t_n)h.$$

A stochastic differential equation has an additional random term:

$$dX = f(X, t) dt + g(X, t) dW.$$

Here,

$$f(X, t)$$

is called the *drift*, while

$$g(X, t)$$

is called the *diffusion coefficient*.

For a finite time-step, the corresponding update is

$$X_{n+1} = X_n + f(X_n, t_n)h + g(X_n, t_n)\Delta W_n,$$

where ΔW_n is a random Brownian increment.

Setting $g = 0$ removes the stochastic term and recovers the deterministic equation.

Worked Example 3.1: Deterministic decay with and without noise

Compare the ODE

$$dy = -2y \, dt$$

with the SDE

$$dX = -2X \, dt + 0.5 \, dW.$$

Suppose

$$y_0 = X_0 = 1, \quad h = 0.1, \quad \Delta W_0 = 0.2.$$

Calculate one numerical update for each equation.

Solution. For the ODE, the drift is

$$f(y, t) = -2y.$$

The deterministic Euler update is

$$y_1 = y_0 + (-2y_0)h.$$

Therefore,

$$y_1 = 1 - 2(1)(0.1) = \boxed{0.8}.$$

For the SDE, the drift and diffusion are

$$f(X, t) = -2X, \quad g(X, t) = 0.5.$$

The stochastic update is

$$X_1 = X_0 + (-2X_0)h + 0.5\Delta W_0.$$

Substituting the values,

$$X_1 = 1 - 2(1)(0.1) + 0.5(0.2).$$

Hence

$$\boxed{X_1 = 0.9}.$$

Both equations have the same deterministic drift contribution:

$$-2(1)(0.1) = -0.2.$$

The SDE also receives the random contribution

$$0.5(0.2) = 0.1.$$

If the Brownian increment had instead been

$$\Delta W_0 = -0.2,$$

the stochastic update would have been

$$X_1 = 1 - 0.2 - 0.1 = 0.7.$$

Thus the ODE produces one fixed next value, whereas the SDE can produce different next values depending on the random increment.

Worked Example 3.2: noisy relaxation towards a resting state

Let $X(t)$ be a neuron's voltage relative to its equilibrium voltage. A deterministic relaxation model is

$$dX = -\frac{X}{\tau} dt.$$

A noisy version is

$$dX = -\frac{X}{\tau} dt + \sigma dW.$$

Let

$$X_0 = -10 \text{ mV}, \quad \tau = 20 \text{ ms}, \quad \sigma = 0.5 \text{ mV ms}^{-1/2}.$$

Use

$$h = 1 \text{ ms}, \quad \Delta W_0 = 0.3 \text{ ms}^{1/2}$$

to calculate the first stochastic update. Compare it with the deterministic update.

Solution. The deterministic drift contribution is

$$-\frac{X_0}{\tau} h = -\frac{-10}{20}(1) = 0.5 \text{ mV}.$$

Therefore, the deterministic update is

$$X_1^{\text{det}} = -10 + 0.5 = \boxed{-9.5 \text{ mV}}.$$

The stochastic contribution is

$$\sigma \Delta W_0 = (0.5)(0.3) = 0.15 \text{ mV}.$$

Thus the stochastic update is

$$X_1 = -10 + 0.5 + 0.15.$$

Therefore,

$$\boxed{X_1 = -9.35 \text{ mV}}.$$

The drift moves the voltage towards its equilibrium value $X = 0$. The positive random increment moves it slightly further in that direction during this particular step.

If instead

$$\Delta W_0 = -0.3 \text{ ms}^{1/2},$$

then

$$X_1 = -10 + 0.5 - 0.15 = -9.65 \text{ mV}.$$

The drift is the same in both cases, but the random contribution changes from one realisation to another.

Practice Questions

Question 3.1: Identifying drift and diffusion

For each equation below:

- identify the drift $f(X, t)$;
- identify the diffusion coefficient $g(X, t)$;
- state whether the noise is additive or multiplicative;
- write down the deterministic equation obtained when the noise is removed.

(a)

$$dX = (3 - X) dt + 0.4 dW.$$

(b)

$$dX = -X dt + 0.2X dW.$$

(c)

$$dX = (t - X) dt.$$

Question 3.2: Following a supplied sequence of random increments

Consider

$$dX = (1 - 2X) dt + 0.3 dW, \quad X_0 = 0.5.$$

Use the time-step

$$h = 0.2$$

and the supplied Brownian increments

$$\Delta W_0 = 0.1, \quad \Delta W_1 = -0.25, \quad \Delta W_2 = 0.05.$$

- Write down the numerical update.
- Calculate X_1 .
- Calculate X_2 .
- Calculate X_3 .
- Explain why repeating the calculation with different random increments would generally produce a different trajectory.

Question 3.3: Mean and variance of a one-step update

Consider

$$dX = aX dt + \sigma dW, \quad X(0) = X_0.$$

Assume

$$\mathbb{E}[\Delta W] = 0, \quad \text{Var}(\Delta W) = h.$$

(a) Show that one numerical step is

$$X_1 = X_0(1 + ah) + \sigma \Delta W.$$

(b) Show that

$$\mathbb{E}[X_1] = X_0(1 + ah).$$

(c) Show that

$$\text{Var}(X_1) = \sigma^2 h.$$

(d) Let

$$a = -1, \quad X_0 = 2, \quad h = 0.1, \quad \sigma = 0.4.$$

Calculate the mean and variance of X_1 .

Question 3.4: Different trajectories from the same initial condition

Consider

$$dX = -X dt + dW, \quad X_0 = 1,$$

with

$$h = 0.25.$$

Two realisations use the following Brownian increments:

	ΔW_0	ΔW_1	ΔW_2
Path A	0.2	-0.1	0.05
Path B	-0.2	0.1	-0.05

(a) Show that the update rule is

$$X_{n+1} = 0.75X_n + \Delta W_n.$$

(b) Calculate X_1 , X_2 and X_3 for Path A.

(c) Calculate X_1 , X_2 and X_3 for Path B.

(d) Calculate the deterministic Euler value after three steps.

(e) Compare the three trajectories.

Question 3.5: Noisy firing-rate relaxation

A firing rate relaxes towards

$$R_\infty = 20 \text{ Hz}$$

according to

$$dR = \frac{R_\infty - R}{\tau} dt + \sigma dW.$$

Let

$$R_0 = 5 \text{ Hz}, \quad \tau = 10 \text{ ms}, \quad \sigma = 1 \text{ Hz ms}^{-1/2}, \quad h = 2 \text{ ms}.$$

Use the increments

$$\Delta W_0 = 0.5 \text{ ms}^{1/2}, \quad \Delta W_1 = -0.3 \text{ ms}^{1/2}, \quad \Delta W_2 = 0.2 \text{ ms}^{1/2}.$$

- Write down the stochastic update.
- Calculate R_1 , R_2 and R_3 .
- Calculate the corresponding deterministic Euler trajectory by setting every Brownian increment equal to zero.
- Compare the stochastic and deterministic values after three steps.
- Explain why the stochastic trajectory does not need to move towards R_∞ during every individual time-step.

3.2 Brownian Motion and Random Increments

A standard Brownian motion $W(t)$ satisfies

$$W(0) = 0.$$

For $s > 0$, its increments satisfy

$$W(t+s) - W(t) \sim \mathcal{N}(0, s).$$

Therefore, an increment over a time-step h has mean zero, variance h , and standard deviation $\sqrt{h}$:

$$\Delta W_n = W(t_{n+1}) - W(t_n) \sim \mathcal{N}(0, h).$$

Equivalently, Brownian increments can be generated using

$$\boxed{\Delta W_n = \sqrt{h} \xi_n}, \quad \xi_n \sim \mathcal{N}(0, 1).$$

Increments over non-overlapping time intervals are independent.

Worked Example 3.3: Constructing a Brownian path from normal samples

Construct a Brownian path over the interval $0 \leq t \leq 1$ using the step size

$$h = 0.25$$

and the supplied standard normal values

$$\xi_0 = 1, \quad \xi_1 = -0.5, \quad \xi_2 = 0.2, \quad \xi_3 = -1.2.$$

Take

$$W(0) = 0.$$

Solution. Since

$$h = 0.25,$$

we have

$$\sqrt{h} = 0.5.$$

The Brownian increments are

$$\Delta W_n = 0.5\xi_n.$$

Therefore,

$$\Delta W_0 = 0.5,$$

$$\Delta W_1 = -0.25,$$

$$\Delta W_2 = 0.1,$$

and

$$\Delta W_3 = -0.6.$$

Beginning with $W_0 = 0$, the first path value is

$$W_1 = W_0 + \Delta W_0 = 0.5.$$

The remaining values are

$$W_2 = 0.5 - 0.25 = 0.25,$$

$$W_3 = 0.25 + 0.1 = 0.35,$$

and

$$W_4 = 0.35 - 0.6 = -0.25.$$

Thus the sample path is

n	t_n	W_n
0	0	0
1	0.25	0.50
2	0.50	0.25
3	0.75	0.35
4	1.00	-0.25

In particular, this realisation gives

$$W(1) = -0.25.$$

Across many independent realisations,

$$W(1) \sim \mathcal{N}(0, 1).$$

The value -0.25 is one possible sample from this distribution.

Worked Example 3.4: noisy evidence accumulation

Suppose a neural decision variable $X(t)$ accumulates evidence according to

$$dX = \mu dt + \sigma dW, \quad X(0) = 0,$$

where

$$\mu = 0.4 \text{ s}^{-1}, \quad \sigma = 0.6 \text{ s}^{-1/2}.$$

- Write down the solution $X(t)$.
- Find the distribution of $X(5)$.
- Find the mean, variance and standard deviation of $X(5)$.

(d) Calculate $X(5)$ for a realisation in which

$$W(5) = -0.8 \text{ s}^{1/2}.$$

Solution. Integrating the SDE gives

$$X(t) = X(0) + \mu t + \sigma W(t).$$

Since $X(0) = 0$,

$$X(t) = 0.4t + 0.6W(t).$$

Because

$$W(5) \sim \mathcal{N}(0, 5),$$

we have

$$0.6W(5) \sim \mathcal{N}(0, 0.6^2(5)).$$

Therefore,

$$X(5) \sim \mathcal{N}(2, 1.8).$$

The mean is

$$\mathbb{E}[X(5)] = 2.$$

The variance is

$$\text{Var}(X(5)) = 1.8,$$

and the standard deviation is

$$\sqrt{1.8} \approx 1.34.$$

For the supplied realisation,

$$X(5) = 0.4(5) + 0.6(-0.8).$$

Thus

$$X(5) = 2 - 0.48,$$

so

$$X(5) = 1.52.$$

The deterministic evidence accumulation would give $X(5) = 2$. In this realisation, the accumulated noise has reduced the decision variable below its deterministic value.

Practice Questions

Question 3.6: Distributions of Brownian increments

Let $W(t)$ be a standard Brownian motion.

For each random variable below, state its mean, variance, standard deviation and probability distribution:

(a)

$$W(3) - W(1).$$

(b)

$$W(3.5) - W(3).$$

(c)

$$W(4) - W(1).$$

(d)

$$2(W(2) - W(1.5)).$$

Explain why the standard deviation of a Brownian increment scales as the square root of the interval length rather than the interval length itself.

Question 3.7: Generating a Brownian sample path

Use the step size

$$h = 0.04$$

and the standard normal values

$$\xi_0 = 0.5, \quad \xi_1 = -1, \quad \xi_2 = 1.5, \quad \xi_3 = 0, \quad \xi_4 = -0.5.$$

(a) Calculate $\sqrt{h}$.

(b) Calculate the five Brownian increments.

(c) Starting from $W(0) = 0$, calculate the Brownian path at

$$t = 0.04, \quad 0.08, \quad 0.12, \quad 0.16, \quad 0.20.$$

(d) State the theoretical distribution of $W(0.20)$.

(e) Compare the final path value with the theoretical mean and standard deviation.

Question 3.8: Adding independent increments

The interval $0 \leq t \leq 2$ is divided into three parts:

$$[0, 0.25], \quad [0.25, 1], \quad [1, 2].$$

Define

$$\Delta W_1 = W(0.25) - W(0),$$

$$\Delta W_2 = W(1) - W(0.25),$$

and

$$\Delta W_3 = W(2) - W(1).$$

(a) State the distribution of each increment.

(b) Explain why the three increments are independent.

(c) Find the distribution of

$$\Delta W_1 + \Delta W_2 + \Delta W_3.$$

- (d) Suppose one realisation gives

$$\Delta W_1 = 0.2, \quad \Delta W_2 = -0.4, \quad \Delta W_3 = 0.7.$$

Calculate $W(2)$.

- (e) Given that $W(1) = 0.3$, state the conditional distribution of $W(2)$.

Question 3.9: Brownian motion with drift

Consider

$$dX = \mu dt + \sigma dW, \quad X(0) = 1,$$

where

$$\mu = -0.2, \quad \sigma = 0.5.$$

- Write down $X(t)$.
- Find the distribution, mean, variance and standard deviation of $X(4)$.
- Find the distribution, mean, variance and standard deviation of $X(9)$.
- Find the mean and variance of

$$X(9) - X(4).$$

- Explain how the drift and diffusion make different contributions to the behaviour of $X(t)$.

Question 3.10: Variability in an evidence accumulator

A neural evidence variable satisfies

$$dA = 0.3 dt + 0.4 dW, \quad A(0) = 0,$$

where time is measured in seconds.

- Find the distribution of $A(10)$.
- Calculate its mean, variance and standard deviation.
- Calculate $A(10)$ for a path with

$$W(10) = 2.$$

- Calculate $A(10)$ for a second path with

$$W(10) = -1.5.$$

- Calculate the ratio

$$\frac{\mathbb{E}[A(10)]}{\sqrt{\text{Var}(A(10))}}.$$

- Repeat the mean and variance calculation if the diffusion coefficient is increased from 0.4 to 0.8.

- (g) Explain what increasing the diffusion coefficient does to the reliability of the accumulated evidence.

3.3 Euler–Maruyama Simulation

Consider the stochastic differential equation

$$dX = a(X, t) dt + b(X, t) dW.$$

The Euler–Maruyama method uses the update

$$X_{n+1} = X_n + a(X_n, t_n)h + b(X_n, t_n)\Delta W_n,$$

where

$$\Delta W_n \sim \mathcal{N}(0, h).$$

Equivalently, write

$$\Delta W_n = \sqrt{h} \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1).$$

The update then becomes

$$X_{n+1} = X_n + a(X_n, t_n)h + b(X_n, t_n)\sqrt{h}\xi_n.$$

If $b = 0$, Euler–Maruyama reduces to the ordinary Euler method.

Worked Example 3.5: Simulating an SDE with additive noise

Consider

$$dX = (1 - X) dt + 0.4 dW, \quad X(0) = 0.$$

Use Euler–Maruyama with

$$h = 0.25$$

and the standard normal values

$$\xi_0 = 0.5, \quad \xi_1 = -1, \quad \xi_2 = 0.25$$

to calculate X_1 , X_2 and X_3 .

Solution. Since

$$\sqrt{h} = \sqrt{0.25} = 0.5,$$

the Euler–Maruyama update is

$$X_{n+1} = X_n + (1 - X_n)(0.25) + 0.4(0.5)\xi_n.$$

Therefore,

$$X_{n+1} = 0.75X_n + 0.25 + 0.2\xi_n.$$

For the first step,

$$X_1 = 0.75(0) + 0.25 + 0.2(0.5).$$

Thus

$$X_1 = 0.35.$$

For the second step,

$$X_2 = 0.75(0.35) + 0.25 + 0.2(-1).$$

Therefore,

$$X_2 = 0.2625 + 0.25 - 0.2,$$

so

$$X_2 = 0.3125.$$

For the third step,

$$X_3 = 0.75(0.3125) + 0.25 + 0.2(0.25).$$

Hence

$$X_3 = 0.234375 + 0.25 + 0.05,$$

giving

$$X_3 = 0.534375.$$

The simulated path is therefore

t_n	X_n
0	0
0.25	0.35
0.50	0.3125
0.75	0.534375

The decrease between the first and second steps occurs because the negative random increment is large enough to oppose the positive deterministic drift.

Worked Example 3.6: a noisy membrane trajectory

Consider the noisy subthreshold membrane equation

$$dV = \frac{V_\infty - V}{\tau_m} dt + \sigma dW,$$

where

$$V_\infty = -45 \text{ mV}, \quad \tau_m = 20 \text{ ms}, \quad \sigma = 1 \text{ mV ms}^{-1/2}.$$

The initial voltage and threshold are

$$V_0 = -65 \text{ mV}, \quad V_{\text{th}} = -50 \text{ mV}.$$

Use

$$h = 4 \text{ ms}$$

and

$$\xi_0 = 0.5, \quad \xi_1 = -1, \quad \xi_2 = 0.25, \quad \xi_3 = 2$$

to simulate four Euler–Maruyama steps. Determine when threshold is first crossed.

Solution. Since

$$\sqrt{h} = \sqrt{4} = 2 \text{ ms}^{1/2},$$

the update is

$$V_{n+1} = V_n + \frac{-45 - V_n}{20}(4) + 2\xi_n.$$

Therefore,

$$V_{n+1} = 0.8V_n - 9 + 2\xi_n.$$

For the first step,

$$V_1 = 0.8(-65) - 9 + 2(0.5) = -60 \text{ mV}.$$

For the second step,

$$V_2 = 0.8(-60) - 9 + 2(-1) = -59 \text{ mV}.$$

For the third step,

$$V_3 = 0.8(-59) - 9 + 2(0.25) = -55.7 \text{ mV}.$$

For the fourth step,

$$V_4 = 0.8(-55.7) - 9 + 2(2) = -49.56 \text{ mV}.$$

Thus

t_n (ms)	V_n (mV)
0	-65
4	-60
8	-59
12	-55.7
16	-49.56

The first numerical value above threshold occurs at

$$t = 16 \text{ ms}.$$

For comparison, the deterministic model reaches threshold at

$$t_{\text{sp}} = 20 \ln 4 \approx 27.7 \text{ ms}.$$

In this particular realisation, positive noise causes an earlier threshold crossing.

Practice Questions

Question 3.11: Euler–Maruyama with additive noise

Consider

$$dX = -2X dt + 0.5 dW, \quad X_0 = 1.$$

Use

$$h = 0.1$$

and

$$\xi_0 = 1, \quad \xi_1 = -0.5, \quad \xi_2 = 0.25.$$

- Calculate $\sqrt{h}$.
- Write down the Euler–Maruyama update.
- Calculate X_1 , X_2 and X_3 .
- State the deterministic Euler update obtained by setting the diffusion coefficient equal to zero.

Question 3.12: Multiplicative noise

Consider

$$dX = 0.5X dt + 0.3X dW, \quad X_0 = 2.$$

Use

$$h = 0.25$$

and

$$\xi_0 = 0.4, \quad \xi_1 = -1, \quad \xi_2 = 0.5.$$

- Explain why the noise is multiplicative.
- Show that the Euler–Maruyama update can be written as

$$X_{n+1} = X_n (1.125 + 0.15\xi_n).$$

- Calculate X_1 , X_2 and X_3 .
- Explain how the size of the random contribution depends on the current value X_n .

Question 3.13: Conditional mean and variance of one step

Consider

$$dX = a(X, t) dt + b(X, t) dW.$$

Suppose that at time t_n ,

$$X_n = x.$$

- Show that the Euler–Maruyama update is

$$X_{n+1} = x + a(x, t_n)h + b(x, t_n)\sqrt{h}\xi_n.$$

- Show that, conditional on $X_n = x$,

$$\mathbb{E}[X_{n+1} \mid X_n = x] = x + a(x, t_n)h.$$

(c) Show that

$$\text{Var}(X_{n+1} \mid X_n = x) = b(x, t_n)^2 h.$$

(d) Apply these results to

$$dX = (1 - 2X) dt + 0.4 dW,$$

when

$$x = 0.5, \quad h = 0.2.$$

Question 3.14: How drift and noise scale with the step size

Consider the SDE

$$dX = a dt + \sigma dW,$$

where a and σ are constants.

Compare two step sizes:

$$h_1 = 0.04, \quad h_2 = 0.01.$$

(a) By what factor does the deterministic drift contribution change when h_1 is replaced by h_2 ?

(b) By what factor does the standard deviation of the random contribution change?

(c) Explain why the correct random contribution is

$$\sigma \sqrt{h} \xi$$

rather than

$$\sigma h \xi.$$

(d) What variance would the incorrect expression $\sigma h \xi$ produce?

Question 3.15: Noisy subthreshold voltage

A membrane voltage satisfies

$$dV = \frac{-50 - V}{10} dt + 0.6 dW,$$

where voltage is measured in millivolts and time in milliseconds.

Let

$$V_0 = -70 \text{ mV}, \quad h = 2 \text{ ms},$$

and use

$$\xi_0 = 0.5, \quad \xi_1 = -1, \quad \xi_2 = 0.25, \quad \xi_3 = 0.$$

(a) Write down the Euler–Maruyama update.

(b) Calculate the voltage at

$$t = 2, \quad 4, \quad 6, \quad 8 \text{ ms}.$$

(c) Calculate the corresponding deterministic Euler trajectory.

(d) Compare the deterministic and stochastic voltages at $t = 8 \text{ ms}$.

- (e) Explain why a single stochastic trajectory is not expected to remain consistently above or below the deterministic trajectory.

3.4 Itô's Formula and Change of Variables

Suppose

$$dX = a(X, t) dt + b(X, t) dW,$$

and define

$$Y = F(X, t).$$

Itô's formula is

$$dY = \left(\frac{\partial F}{\partial t} + a \frac{\partial F}{\partial X} + \frac{1}{2} b^2 \frac{\partial^2 F}{\partial X^2} \right) dt + b \frac{\partial F}{\partial X} dW.$$

If F does not depend explicitly on time, this reduces to

$$dY = \left(aF'(X) + \frac{1}{2} b^2 F''(X) \right) dt + bF'(X) dW.$$

The term

$$\frac{1}{2} b^2 F''(X) dt$$

is the *Itô correction*. It is absent from the ordinary chain rule.

Worked Example 3.7: Geometric Brownian motion

Suppose $X(t) > 0$ satisfies

$$dX = \mu X dt + \sigma X dW, \quad X(0) = X_0 > 0.$$

Define

$$Y = \ln X.$$

Use Itô's formula to find the SDE for Y , solve it, and hence find $X(t)$.

Solution. For

$$F(X) = \ln X,$$

the derivatives are

$$F'(X) = \frac{1}{X}$$

and

$$F''(X) = -\frac{1}{X^2}.$$

The drift and diffusion coefficients of X are

$$a(X, t) = \mu X, \quad b(X, t) = \sigma X.$$

Itô's formula gives

$$dY = \left[\mu X \left(\frac{1}{X} \right) + \frac{1}{2} \sigma^2 X^2 \left(-\frac{1}{X^2} \right) \right] dt + \sigma X \left(\frac{1}{X} \right) dW.$$

Therefore,

$$dY = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW.$$

Integrating from 0 to t ,

$$Y(t) = Y(0) + \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W(t).$$

Since

$$Y(0) = \ln X_0,$$

we have

$$\ln X(t) = \ln X_0 + \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W(t).$$

Exponentiating,

$$X(t) = X_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W(t) \right].$$

The ordinary chain rule would incorrectly give a logarithmic drift of μ . Itô's formula gives the corrected drift

$$\mu - \frac{\sigma^2}{2}.$$

Worked Example 3.8: mean-square membrane fluctuations

Let $X(t)$ represent a membrane-potential deviation from equilibrium. Suppose

$$dX = -\frac{X}{\tau} dt + \sigma dW, \quad X(0) = 0.$$

Use Itô's formula with

$$Y = X^2$$

to find an equation for

$$M(t) = \mathbb{E}[X(t)^2].$$

Hence find the long-term mean-square fluctuation. Evaluate the result when

$$\tau = 10 \text{ ms}, \quad \sigma = 0.4 \text{ mV ms}^{-1/2}.$$

Solution. Let

$$F(X) = X^2.$$

Then

$$F'(X) = 2X, \quad F''(X) = 2.$$

The drift and diffusion coefficients are

$$a(X, t) = -\frac{X}{\tau}, \quad b(X, t) = \sigma.$$

Itô's formula gives

$$d(X^2) = \left[-\frac{X}{\tau}(2X) + \frac{1}{2}\sigma^2(2) \right] dt + \sigma(2X) dW.$$

Therefore,

$$d(X^2) = \left(-\frac{2X^2}{\tau} + \sigma^2 \right) dt + 2\sigma X dW.$$

Taking expectations, the stochastic integral has mean zero. Thus

$$\frac{dM}{dt} = -\frac{2M}{\tau} + \sigma^2.$$

Since $X(0) = 0$,

$$M(0) = 0.$$

Solving this first-order linear equation gives

$$M(t) = \frac{\sigma^2 \tau}{2} \left(1 - e^{-2t/\tau} \right).$$

As t becomes large,

$$M(t) \rightarrow \frac{\sigma^2 \tau}{2}.$$

For the stated parameters,

$$\frac{\sigma^2 \tau}{2} = \frac{(0.4)^2(10)}{2} = 0.8 \text{ mV}^2.$$

Therefore, the long-term mean-square fluctuation is

$$0.8 \text{ mV}^2.$$

At $t = 10 \text{ ms}$,

$$M(10) = 0.8(1 - e^{-2}) \approx 0.692 \text{ mV}^2.$$

Because the mean of $X(t)$ is zero, $M(t)$ is also its variance. The root-mean-square fluctuation at 10 ms is

$$\sqrt{M(10)} \approx 0.832 \text{ mV}.$$

Practice Questions

Question 3.16: A quadratic transformation

Suppose

$$dX = (2 - X) dt + 0.5 dW.$$

Define

$$Y = X^2.$$

- (a) Calculate $F'(X)$ and $F''(X)$ for $F(X) = X^2$.
- (b) Use Itô's formula to find the SDE satisfied by Y .
- (c) Identify the Itô correction term.
- (d) When $X = 1$, calculate the instantaneous drift and diffusion coefficients of Y .

Question 3.17: An exponential transformation

Suppose

$$dX = \mu dt + \sigma dW$$

and define

$$Y = e^X.$$

- (a) Use Itô's formula to show that

$$dY = \left(\mu + \frac{\sigma^2}{2} \right) Y dt + \sigma Y dW.$$

- (b) Explain why the noise in the transformed equation is multiplicative.
- (c) State the result that would be obtained by incorrectly using the ordinary chain rule.
- (d) Let

$$\mu = -0.2, \quad \sigma = 0.6.$$

Calculate the corrected drift coefficient of Y .

Question 3.18: The reciprocal of geometric Brownian motion

Suppose

$$dX = \mu X dt + \sigma X dW, \quad X > 0.$$

Define

$$Y = \frac{1}{X}.$$

- (a) Calculate $F'(X)$ and $F''(X)$ for $F(X) = X^{-1}$.
- (b) Use Itô's formula to show that

$$dY = (\sigma^2 - \mu)Y dt - \sigma Y dW.$$

- (c) Let

$$\mu = 0.1, \quad \sigma = 0.4.$$

Write the transformed SDE numerically.

- (d) Explain why taking a reciprocal changes both the drift and the sign of the diffusion term.

Question 3.19: A time-dependent transformation

Suppose

$$dX = -X dt + dW$$

and define

$$Y = tX^2.$$

- (a) For

$$F(X, t) = tX^2,$$

calculate

$$\frac{\partial F}{\partial t}, \quad \frac{\partial F}{\partial X}, \quad \frac{\partial^2 F}{\partial X^2}.$$

- (b) Use the time-dependent form of Itô's formula to show that

$$dY = [(1 - 2t)X^2 + t] dt + 2tX dW.$$

- (c) Identify separately the contribution from explicit time dependence and the Itô correction.

Question 3.20: Voltage variance under noisy relaxation

A membrane-potential deviation satisfies

$$dV = -\frac{V}{\tau} dt + \sigma dW, \quad V(0) = V_0.$$

- (a) Apply Itô's formula to V^2 .

- (b) Show that

$$M(t) = \mathbb{E}[V(t)^2]$$

satisfies

$$\frac{dM}{dt} = -\frac{2M}{\tau} + \sigma^2.$$

- (c) Show that

$$M(t) = \frac{\sigma^2 \tau}{2} + \left(V_0^2 - \frac{\sigma^2 \tau}{2} \right) e^{-2t/\tau}.$$

- (d) Show that

$$\mathbb{E}[V(t)] = V_0 e^{-t/\tau}.$$

- (e) Hence show that

$$\text{Var}(V(t)) = \frac{\sigma^2 \tau}{2} (1 - e^{-2t/\tau}).$$

- (f) Let

$$V_0 = 3 \text{ mV}, \quad \tau = 20 \text{ ms}, \quad \sigma = 0.5 \text{ mV ms}^{-1/2}.$$

Calculate the mean, second moment, variance and standard deviation at $t = 20 \text{ ms}$.

3.5 The Stochastic Leaky Integrate-and-Fire Model

A stochastic leaky integrate-and-fire model can be written as

$$dV = \frac{V_\infty - V}{\tau_m} dt + \sigma dW,$$

where

$$V_\infty = E_L + R_m I_0.$$

The deterministic drift pulls the voltage towards V_∞ , while the Brownian term introduces random voltage fluctuations.

When the voltage reaches the threshold V_{th} , a spike is recorded and the voltage is reset:

$$V \longrightarrow V_{\text{reset}}.$$

For a numerical time-step h , the Euler–Maruyama candidate update is

$$\tilde{V}_{n+1} = V_n + \frac{V_\infty - V_n}{\tau_m} h + \sigma \sqrt{h} \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1).$$

The threshold and reset rule is then

$$V_{n+1} = \begin{cases} V_{\text{reset}}, & \tilde{V}_{n+1} \geq V_{\text{th}}, \\ \tilde{V}_{n+1}, & \tilde{V}_{n+1} < V_{\text{th}}. \end{cases}$$

Worked Example 3.9: Mean and variance of the subthreshold voltage

Consider

$$dV = \frac{V_\infty - V}{\tau_m} dt + \sigma dW, \quad V(0) = V_0,$$

before any threshold or reset is applied.

Show that

$$\mathbb{E}[V(t)] = V_\infty + (V_0 - V_\infty)e^{-t/\tau_m}$$

and

$$\text{Var}(V(t)) = \frac{\sigma^2 \tau_m}{2} \left(1 - e^{-2t/\tau_m}\right).$$

Evaluate these quantities at $t = 20$ ms when

$$V_\infty = -55 \text{ mV}, \quad V_0 = -65 \text{ mV}, \quad \tau_m = 20 \text{ ms},$$

and

$$\sigma = 0.5 \text{ mV ms}^{-1/2}.$$

Solution. Let

$$m(t) = \mathbb{E}[V(t)].$$

Taking expectations in the SDE gives

$$\frac{dm}{dt} = \frac{V_\infty - m}{\tau_m},$$

because the Brownian term has mean zero.

Therefore,

$$m(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau_m}.$$

The voltage fluctuations around the mean have variance

$$\text{Var}(V(t)) = \frac{\sigma^2 \tau_m}{2} (1 - e^{-2t/\tau_m}).$$

At $t = 20$ ms,

$$m(20) = -55 + (-65 + 55)e^{-1}.$$

Thus

$$m(20) = -55 - 10e^{-1} \approx -58.68 \text{ mV}.$$

The variance is

$$\text{Var}(V(20)) = \frac{(0.5)^2(20)}{2} (1 - e^{-2}).$$

Therefore,

$$\text{Var}(V(20)) \approx 2.16 \text{ mV}^2.$$

The standard deviation is

$$\text{SD}(V(20)) \approx 1.47 \text{ mV}.$$

As t becomes large,

$$\mathbb{E}[V(t)] \rightarrow V_\infty$$

and

$$\text{Var}(V(t)) \rightarrow \frac{\sigma^2 \tau_m}{2}.$$

These formulas describe the subthreshold process. Once threshold and reset are imposed, the full voltage distribution is no longer given by this unrestricted Gaussian process.

Worked Example 3.10: Simulating a spike and reset

Consider

$$dV = \frac{-50 - V}{10} dt + dW,$$

where voltage is measured in millivolts and time in milliseconds.

Let

$$V_0 = -65 \text{ mV}, \quad V_{\text{th}} = -55 \text{ mV}, \quad V_{\text{reset}} = -65 \text{ mV}.$$

Use Euler–Maruyama with

$$h = 2 \text{ ms}$$

and

$$\xi_0 = 0.5, \quad \xi_1 = 1, \quad \xi_2 = 1.5, \quad \xi_3 = -0.5, \quad \xi_4 = 1.2.$$

Simulate five steps, applying the reset immediately after a threshold crossing.

Solution. Since

$$\sqrt{h} = \sqrt{2},$$

the candidate update is

$$\tilde{V}_{n+1} = V_n + \frac{-50 - V_n}{10}(2) + \sqrt{2}\xi_n.$$

Equivalently,

$$\tilde{V}_{n+1} = 0.8V_n - 10 + \sqrt{2}\xi_n.$$

For the first step,

$$\tilde{V}_1 = 0.8(-65) - 10 + \sqrt{2}(0.5) \approx -61.29 \text{ mV}.$$

This is below threshold, so

$$V_1 = -61.29 \text{ mV}.$$

The subsequent calculations are

t (ms)	ξ_n	$\tilde{V}_{n+1}$ (mV)	V_{n+1} (mV)
2	0.5	-61.29	-61.29
4	1.0	-57.62	-57.62
6	1.5	-53.97	-65.00
8	-0.5	-62.71	-62.71
10	1.2	-58.47	-58.47

At $t = 6$ ms, the candidate voltage satisfies

$$\tilde{V}_3 \geq V_{\text{th}}.$$

A spike is therefore recorded, and the stored voltage is reset to

$$V_3 = V_{\text{reset}} = -65 \text{ mV}.$$

Thus the simulated spike time is

$$t_{\text{sp}} = 6 \text{ ms}.$$

Practice Questions

Question 3.21: Components of the stochastic LIF model

Consider

$$dV = \frac{V_{\infty} - V}{\tau_m} dt + \sigma dW.$$

- Identify the drift coefficient.
- Identify the diffusion coefficient.
- Write down the deterministic LIF equation obtained when $\sigma = 0$.
- State the deterministic equilibrium voltage.
- Write down the Euler–Maruyama update using a time-step h and standard normal variables ξ_n .
- Explain the roles of τ_m and σ .

Question 3.22: Threshold crossing and reset

Consider

$$dV = \frac{-48 - V}{12} dt + 0.8 dW,$$

with

$$V_0 = -66 \text{ mV}, \quad V_{\text{th}} = -54 \text{ mV}, \quad V_{\text{reset}} = -64 \text{ mV}.$$

Use

$$h = 3 \text{ ms}$$

and

$$\xi_0 = 0.4, \quad \xi_1 = -0.6, \quad \xi_2 = 1.2, \quad \xi_3 = 0.8, \quad \xi_4 = -0.5.$$

- Write down the Euler–Maruyama candidate update.
- Calculate the five candidate voltages.
- Apply the threshold and reset rule.
- State the first numerical spike time.
- State the stored voltage after the final step.

Question 3.23: Subthreshold voltage statistics

Before threshold and reset are imposed, suppose

$$dV = \frac{V_{\infty} - V}{\tau_m} dt + \sigma dW, \quad V(0) = V_0.$$

Let

$$V_{\infty} = -60 \text{ mV}, \quad V_0 = -70 \text{ mV}, \quad \tau_m = 25 \text{ ms},$$

and

$$\sigma = 0.4 \text{ mV ms}^{-1/2}.$$

- Calculate $\mathbb{E}[V(25 \text{ ms})]$.
- Calculate $\text{Var}(V(25 \text{ ms}))$.
- Calculate the corresponding standard deviation.
- Find the limiting mean and variance as $t \rightarrow \infty$.
- Explain how the limiting variance changes if σ is doubled.

Question 3.24: Probability of crossing threshold in one step

At the beginning of a numerical step, suppose

$$V_n = -56 \text{ mV}.$$

The stochastic LIF parameters are

$$V_{\infty} = -60 \text{ mV}, \quad \tau_m = 20 \text{ ms}, \quad \sigma = 2 \text{ mV ms}^{-1/2},$$

with

$$h = 1 \text{ ms}$$

and

$$V_{\text{th}} = -55 \text{ mV}.$$

- (a) Find the conditional mean of the candidate voltage $\tilde{V}_{n+1}$.
- (b) Find its conditional variance and standard deviation.
- (c) Show that

$$\mathbb{P}\left(\tilde{V}_{n+1} \geq V_{\text{th}} \mid V_n = -56\right) = 1 - \Phi(0.6),$$

where Φ is the standard normal cumulative distribution function.

- (d) Use

$$\Phi(0.6) \approx 0.726$$

to estimate the crossing probability.

- (e) Explain why threshold crossing is possible even though the deterministic drift points away from threshold.

Question 3.25: Summarising stochastic spike-time variability

Five simulated interspike intervals from a stochastic LIF neuron are

$$18, \quad 22, \quad 20, \quad 25, \quad 15 \text{ ms}.$$

- (a) Calculate the mean interspike interval.
- (b) Calculate the mean firing rate in hertz using

$$f = \frac{1000}{\text{mean interval in ms}}.$$

- (c) Using division by 5, calculate the standard deviation of the five intervals.
- (d) Calculate the coefficient of variation

$$\text{CV} = \frac{\text{standard deviation}}{\text{mean}}.$$

- (e) A deterministic LIF model with the same constant input has an interspike interval of exactly 20 ms. Compare its mean firing rate and spike-time variability with the stochastic model.